



MASINDE MULIRO UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL BASED PROGRAMME

PSY 311: EDUCATIONAL MEASUREMENT AND EVALUATION

FACULTY OF EDUCATION AND SOCIAL SCIENCES

DEPARTMENT OF EDUCATIONAL PSYCHOLOGY

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SYMBOLS

S	– Sum of
<i>f</i>	– Frequencies
<i>N</i> or <i>n</i>	– Number of variables
<i>Mo</i>	– Mode
<i>Md</i>	– Median

Introduction to the Module



This is PSY 311: Educational Measurement and Evaluation Module. This is a 3rd Year, Second Semester Module. It is our belief that you were introduced to PSY 210 and PSY 310, both of which made several mention of measurement and evaluation aspects in psychological testing.

As you read through this module, you will be introduced terminologies used in measurement and evaluation, the importance of measurement and evaluation, types of measurement and evaluation, construction of tests and their administration. You will also learn how to prepare a frequency table from raw data, measures of central tendency, measures of dispersion/variability, measures of relationship, and prediction of outcomes based on students' scores.

This module has six major topics and each topic has several sub-topics. Every user of this module has to ensure that before he/she proceeds to a new section, each preceding sub-section is thoroughly comprehended. Each of the sub-section presents self-check tests meant to help you assess your level of understanding. The score earned should tell you the progress you have made in internalizing the information. It is our sincere hope that you will find the module easy to understand and informative. However, should you have any comments or compliments, feel free to do so.

Aim

Module PSY 311 aims at equipping you with knowledge and skills in test measurement and test evaluation and various ways of test interpretation.

Objectives

By the end of the Module, you should be able to:

- i. Define various statistical concepts and explain their importance in educational measurement and evaluation

- ii. Explain and construct different types of tests.
- iii. Tabulate and depict sets of data for both ungrouped and grouped distributions.
- iv. Explain and compute measures of central tendency, variability and relationship.
- v. Explain regression analysis and interpret the standard error of estimate.
- vi. Explain and compute the validity and reliability of a test.

This Module consists of six topics namely;

Topic 1: Test Measurement and Evaluation

Topic 2: Frequency Distributions and Graphic Presentations

Topic 3: Measures of Central Tendency

Topic 4: Measures of Dispersion

Topic 5: Measures of Relationship

Topic 6: Test validity and Reliability

Welcome to PSY 311 Educational Measurement and Evaluation Module

TOPIC 1

TESTS MEASUREMENT AND EVALUATION

1.0 Introduction



In this topic, you will learn types of evaluation, types of tests and examinations, construction of tests, scoring of tests and test administration.

1.1 Objectives



By the end of the topic, you should be able to:

- 👤 Define the terms measurement, evaluation and assessment.
- 👤 State and explain the different types of evaluation and assessment.
- 👤 Explain purposes of measurement and evaluation
- 👤 Describe various types of tests and examination.
- 👤 Explain factors to consider when constructing and scoring a test
- 👤 Explain causes, methods and effects of examination cheating

1.2 Sub-sections of Topic 1

Section 1: Introduction.....

Section 2: Measurement, evaluation and assessment

Section 3: Purposes of Measurement and Evaluation

Section 4: Tests and Examinations

Section 5: Construction of Tests

Section 6: Test Scoring

Section 7: Test/Examination Administration and Examination Cheating

Let us look at each of these sections in detail.

1.3 MEASUREMENT, EVALUATION AND ASSESSMENT

Definitions of terms

- Measurement - is the process of assigning a quantitative value (numerical) to a student's attainment in a given area of learning e.g. 64%.

- Evaluation – refers to the process of assigning a qualitative value to a student's attainment in a given area of learning e.g. C+.

Types of Evaluation

There are three types

1. Formative evaluation
2. Summative evaluation
3. Assessment

Formative Evaluation

- It is the progressive assessment of the success with which a program is being implemented. It shows whether learning objectives are being achieved.
- It is done with a small group of people to "test run" various aspects of instructional materials.
- It is typically conducted during the development or improvement of a program and it is conducted more than once.
- The purpose of formative evaluation is to validate or ensure that the goals of the instruction are being achieved and to improve the instruction, if necessary, by means of identification and subsequent remediation of problematic aspects.
- Formative evaluation is research-oriented.
- Formative evaluation provides information on the product's efficacy (its ability to do what it was designed to do).

Summative Evaluation

- † Summative evaluation is a method of judging the worth of a program at the end of the program activities. The focus is on the outcome.
- † It is typically quantitative and uses numeric scores or letter grades to assess learner achievement.

- It is action-oriented. That is, on the basis of the findings, the programme can be adopted entirely, modified or abandoned altogether.

Assessment

- It is the process by which the quality of an individual's work or performance is judged.
- It is carried out through observations of pupils' at work or by various kinds of tests given periodically.
- When practiced as an ongoing process, such assessment is called continuous assessment.



In a group of five, discuss with specific examples from your school settings the different types of evaluations carried out.

Types of Assessment

1. Normative Assessment/Testing

- It is also called Norm-referenced assessment/test. It is where the quality of the grade depends on the average (norms) performance i.e. an individual's score is judged in relation to how good the overall performance is or was.
- It is not measured against defined criteria but is relative to the student body undertaking the assessment i.e. it will tell you how a child compares to similar children on a given set of skills and knowledge.
- The IQ test is the best known example of norm-referenced assessment. Many entrance tests (to prestigious schools or universities) are norm-referenced e.g. KCPE or KCSE.
- It is a way of comparing students implying that standards may vary from year to year, depending on the quality of the cohort.

Advantages

- i. It does not enforce any expectation of what all students should know or be able to do other than what students can actually demonstrate.
- ii. Present levels of performance and inequity are taken as fact but not as defects to be removed by a redesigned system.
- iii. Aims of student performance are not raised every year until all are proficient. Scores are not required to show continuous improvement.

Limitations

- (a) It cannot measure progress of the population of a whole, only where individuals fall within the whole.
- (b) It does not set what an individual should profess to prove a mastery of a skill being tested but rather bases on the set norm.
- (c) It judges set benchmarks around items of varying difficulty without considering the ability level or age of the examinees.
- (d) The difficulty level of items that determine the levels passing vary from year to year.

2. Criterion Assessment

- It is where a decision is made as to whether a pupil has actually achieved specified level of learning regardless of the performance of other pupils.
- Here, the criterion or level of achievement which warrants a mastery of certain skills is set in advance. It is not flexible.
- Criterion-referenced assessment is often, but not always, used to establish a person's competence in doing something e.g. the driving test, when learner drivers are measured against a range of explicit criteria.
- It tells where the person stands in some population of persons who have taken the test.

- Most criterion-referenced tests involve a cut score, where the examinee passes if their score exceeds the cut score and fails if it does not (often called a mastery test).
- However, not all criterion-referenced tests have a cut score, and the score can simply refer to a person's standing on the subject domain.

Advantage

- i. Many criterion-referenced tests are *high-stakes tests* since results of the test have serious implications for the individual examinee.
- ii. Criterion referenced tests are standard-based assessments where students are assessed with regards to set standards that define what they "should" know.

Limitations

- (a) They can be described as, "you lose a lot if you fail to pass" e.g. licensure testing where the test must be passed in order to progress.
- (b) Some tests set a standard that have failed 50 to 80 percent of students at the outset, a higher, not lower failure rate than is possible with standard definition of 50 percent falling below average.

3. Diagnostic Assessment

It is the process of finding out the exact nature of a person's problem or difficulties. In education, the aim is to give relevant remedial teaching to those who deserve it.

? What is your major teaching subject? Have you ever made diagnostic assessment of your pupils in the subject? What were your major findings?

1.4 PURPOSES OF MEASUREMENT AND EVALUATION

The primary purpose of assessment is to improve student learning.

1. To identify areas of weakness in learning..
2. Helps build a shared understanding of the progress made by pupils in order to provide pointers for further development
3. Provide feedback to students, staff and parents/guardians on pupils' progress and achievements.
4. Timely feedback improves motivation and achievement for the learner.
5. To grade students for purposes of promotion to next level.
6. Acts as a quality assurance mechanism both for internal and external systems i.e. tells whether objectives are been achieved.
7. To appraise the effectiveness of a teaching method or methods.
8. To measure specific abilities e.g. IQ, vocabulary, creativity etc.
9. To provide information for effective educational and vocational Counselling.

? What are the main shortcomings of regional evaluation tests your school participates in?

1.5 TESTS AND EXAMINATIONS

Test - Is a set of questions to which an examinee has to respond.

Examination - Is a set of tests in various areas to which an examinee has to respond.

Types of Examinations

A. Internal Examination

It is usually prepared and marked by the teacher's in-charge of the subject in question.

Advantages

- i. Questions asked are based on the work covered in class and are therefore learner friendly.
- ii. The language and format used in setting the questions are familiar to the learners hence learners experience less stress compared to external examinations.

Disadvantage

- i. The results may not be a true reflection of the learners' ability since the teacher tends to be subjective in his/her evaluation of the learners' performance.
- ii. Teacher may set the questions based on what has been covered in class hence syllabus coverage is poor.
- iii. Tends to be highly subjective since the setter (teacher) sets based on certain preferences.

B. External examination

Is prepared and marked by a person or body of experts not responsible for teaching the subject being examined.(subject in question)

Advantages

- i. It gives a more objective assessment of the learner since the examiners are unknown to the examinee.
- ii. There is good syllabus coverage since both the teacher and the learner cannot guess the examinable areas.
- iii. Due to objectivity in scoring of examinees abilities across the population, higher institutions of learning and potential employers prefer selection on this basis.

Disadvantages

- i. It invalidates the importance of learning and education since it often turns out examination oriented.
- ii. Encourages cramming of facts rather than application of learned materials.
- iii. It increases emotional stress due to over concern about examinations results.

TYPES OF TESTS

A. Objectives Tests

Are questions that demand answers that are either right or wrong and for each of which there is only one possible correct answer.

Advantages

- 1. Are easy to mark and grade.
- 2. Examine a wide coverage of the topics learned hence students read widely.
- 3. They are practical and handy for relatively large classes.
- 4. Human error, bias or prejudice by the marker is removed i.e. scoring is extremely reliable.
- 5. If well set, they have a strong discriminative power between the bright and weak students.
- 6. Learners obtain feedback on their performance much faster.

Disadvantages

- i) Are difficult to set and therefore time consuming.
- ii) They are open to guesswork.
- iii) They limit the learner's use of his/her acquired writing and literary skills e.g. creativity, analysis or evaluation.
- iv) They are relatively expensive in terms of materials needed to produce a complete test.
- v) The selection of questions may greatly be influenced by the examiner's bias.

? Are objective types of tests ideal for use in lower primary schools? Give reasons for your choice of answer?

TYPES OF OBJECTIVE TESTS

There are three categories

- 1) Supply items
- 2) Selection items
- 3) Rank order items

1) Supply items

They are also called completion items. These types of tests require a student to recall or recognize the appropriate term, concept or phrase or to complete a statement.

Sub-types of supply item tests

- a) Filling in blanks
- b) One word answer
- c) Information for maps, diagram's and pictures
- d) Practical experiments.

2) Selection Items

Require a student to choose one alternative from a range of alternatives.

Sub-types of selection item tests

- a) True /False of Right Wrong of Yes/No
- b) Matching pairs
- c) Multiple choice

3) Rank Order Items

Require a student to indicate the appropriate order (serial, chronological, logical etc.) of the items presented.

B. SUBJECTIVE TYPE TESTS

There are three types namely;

- i) Essay type questions
- ii) Assessment of practical skills by observation
- iii) Projects

i) Essay type questions

Are questions that require the candidate to supply a single word or a sentence as an answer.

Advantages

- a) Are easy to set hence time saving.
- b) They are ideal for smaller classes.
- c) Are easier to predict.
- d) Give room for choice and self-expression i.e. have allowance for creativity.
- e) Test both the learner's understanding and insight.
- f) Enable the examiner to follow up answers that may not be clear.
- g) Minimize the extent to which a student can use guesswork to find an answer.

Disadvantages

- a) Permit and occasionally allow bluffing and cheating.
- b) They are tedious and difficult to mark and score
- c) Encourage rote memorization as such may not show the extent to which a learner can apply acquired knowledge.

- d) Since most questions test knowledge aspect, learner's higher level of thinking is often ignored.
- e) Scoring tends to be more subjective rather than objective.
- f) There is incomplete sampling of candidates' knowledge due to limited areas of testing.
- g) Do not adequately predict future academic performance because success sometimes depends on a candidate ability to predict possible exam questions.



Are subjective types of tests suitable for general testing at lower levels of primary schools? Support your argument.

Types of Essay Type Questions

- a) Restricted- response type

Directed the learner as to what information is required.

- b) Extended/open-response type

It has no restrictions as to what or how much to write i.e. examinee has a free hand to write as much on a particular subject.

- 2) Assessment of practical skills by observation.

Requires the examiner needs to identify what particular features of the pupils' performance will be awarded points and what proportion of the total will be allocated to each of these e.g. teaching practice, micro-teaching/counselling, home science practical, art work etc..

- 3). Projects

It is where a study of some particular or local theme is carried out by an individual student and a write up made e.g. the PORTFOLIO in CIT 299.



Think of any practical assessment test you have given to your pupils. What aspects of the practical test were scored?

Other Types of Tests

1. Intelligence tests.

Measure various mental skills considered relevant to intelligence in order to find the Intelligence quotient (IQ) of a child.

2. Diagnostic tests

Seek to identify critical weakness in basic education skills for possible remedial action.

3. Achievement tests

Measure a child's ability in a specific skill in relation to a norm.

4. Personality tests

Help to identify the dominant trait of a child so as to classify him/her personality and provide the kind of learning patterns best suited for him/her.

5. Aptitude tests

Measure specific abilities considered important for a particular task or role.



Tests can fall in any of the following categories.

- a) Closed-books tests

Are tests which do not allow the examinee to make reference on any external material(s). The examinee is expected to remember the information off head.

- b) Open-book tests

Here examinees are allowed to use and apply information that they can find in resource materials e.g. common in language tests.

- c) Take-home tests

The examinee is required to make use of community resources such as the library or any other source of information.



Why are closed-books tests not commonly used in primary and secondary school tests and examinations?

1.6 CONSTRUCTION OF TESTS

Qualities of a Good Test

1. Validity

A good test should measure what it is supposed to measure i.e. it should measure specific objective(s) of the test set. A test that is set in a language that is not understandable is invalid.

2. Reliability

A good test should yield the same results on a re-test on the same group of learners under similar conditions.

3. Practicality /Usability

A test is said to be practical or usable if it can be readily used by the teacher in everyday classroom conditions.

A test which costs too much material to produce or a marking scheme which is hard to make renders a test useless.

? Suggest how a teacher can ensure that a test is valid, reliable and usable?

Factors to consider when constructing a Test

1. Specification of objectives

The kind of vocabularies used should elicit the kind of responses required from the candidates.

2. Content

The examiner should ensure that questions set cover all topics taught/covered in class.

3. Emphasized content areas.

Some content areas/topics should be given more emphasis than others depending on the time spent to cover and the total number of questions usually set from such topics.

4. Ability level of students

Questions set should be able to differentiate between bright, average and weak pupils.

5. Specification for types of domains to be measured.

Questions set should include cognitive, affective and psychomotor domains.

6. Specification of the cognitive domain to be measured.

This include (Bloom's taxonomy)

- a. Knowledge –ability to recall facts
- b. Comprehension –ability to retell a story or given information in own words.
- c. Application –ability to use newly learnt facts in novel situations.
- d. Analysis –ability to break down material from component parts e.g. narrating a story based on a series of pictures.
- e. Synthesis -
- f. Evaluation –ability to judge the value or worth of a given piece of information.

7. Specification Table or Grid Matrix or Test Matrix.

It shows the number of questions from a certain content area. It also shows the cognitive domain to test and the number of items to be set from each cognitive domain.

A Test Matrix for ----- Test for a Std ----- Class

CONTENT AREAS	COGNITIVE DOMAINS						Total
	Know	Compr	Appl	Analy	Synth	Eval	
TOTAL							Grand Total



Importance of specification table

- a) Helps to improve the content validity i.e. gives a balanced test.
- b) Helps a teacher not to concentrate on a particular domain of objectives
- c) Helps in accountability of education i.e. how correct or valid a test measurement is.



Prepare a test matrix in your area of specialization. Does it meet the above standards?

8. Format of test items

A test could be oral or written. A written test is better than oral since it also tests a learner's understanding of the concept being tested. The examiner also needs to decide before hand whether essay or objectives test items will be used.

Essay items are preferred if testing on the higher cognitive objectives while objective items are suitable if testing for knowledge and comprehension.

9. Number of test items

The number of test items to be included in the test must be clearly stated. However, this depends on:

- (i) Items allocated for the test.
- (ii) Types of items chosen i.e. objectives or essay.
- (iii) Complexity of test items and thought process involved.

10. Specification of time limits

Time given for a particular test depends on the mental processes involved and the kind of item format used. For multiple choice item, 45-60 seconds is recommended; complicated mathematics problem or complex reading selection may require 4-5minutes while vocabulary items may take 10-15 seconds.

11. Writing the test items

The examiner should have a thorough grasp of the subject matter dealt with in the test. The setter's qualifications should be indicated. A single writer may be assigned a particular area or have several writers assigned to one cell.

1.7.1 Construction of Objective Test Items

A. Completion Test (Filling in Blanks)

† Completion test requires recall and thinking ability. In this type of test, sentences are presented from which certain words or phrases have been omitted.

† To construct completion items, the following suggestions should be considered.

- i. Instructions should be brief and clear.
- ii. Rephrase text books sentences or paragraphs to avoid rote memorization.
- iii. Do not have too many blanks in a short sentence. Blanks should be placed either at the beginning, near the end, or at the end of a statement.
- iv. Blanks should be of standard length to avoid clues about the length of the completing word.
- v. Always specify in what unit or value a numerical answer should be given.
- vi. Use phrases rather than words to avoid ambiguous responses/answers and allow objective marking.
- vii. Guard against clues that may give away the answers by ensuring that completions do not depend on text book expressions or grammatical form.
- viii. Avoid long and winding statements as they tend to lose meaning and confuse pupils unless well framed.

B. Matching Item Tests

† This consists of two columns, the premises (problem to be answered) and the responses (answers). The examinee needs to make some association between each premises and each response.

The following suggestions need to be taken into consideration when constructing matching items

- i. Do not have too many items on the list. A minimum of 5 and a maximum of 7 is preferred.
- ii. The responses should be more than the premises in order to reduce correct item matching by elimination process.
- iii. Materials selected should be from the same subject so that a given premise has several possible matches in the responses.
- iv. Names should be arranged in an alphabetical order while dates and numbers in sequence. This saves the examinees' time.
- v. Watch for irrelevant but revealing association (clues) which may give away the matching such as singulars and plurals.



Prepare a matching item test based on the following information: African countries against their heads of government.

C. True-False Items

Yes/No; Right/Wrong; + (Plus) or – (Minus) or Positive/Negative can also be used in the place of true/false. To construct true/false items, consider the following suggestions:

- i. Place the symbol “T” and “F” before each question. This will save time when marking.
- ii. The number of true statements should equal those of false statements.
- iii. When arranging the items, avoid any form of pattern of true and false answers.
- iv. Do not use words which will provide clues or hints as this may give away the answer.
- v. Use statements which are absolutely true or false and avoid items which express opinions or which are trivial/tricky.
- vi. Avoid the use of double negatives and single negatives should be used sparingly. However, if they must be used, they should be underlined, capitalized or italicized.
- vii. Do not lift statements/quotations from textbooks since they encourage rote memory and turn out ambiguous when interpreted out of context.



Construct 10 True-False item test for your class taking into account the above suggestions.

D. Multiple-Choice or Best-Answer Items

A multiple-choice test consists of two parts, the stem and a list of suggested answers.

† **The stem:** Contains the statement, questions, phrase or word i.e. the problem part. The stem may be stated as a direct question or as an incomplete statement

† **A list of suggested answers:** The correct answer is called the **key** while the incorrect responses are called **distracters** or **foils**.

Types of multiple choice questions

a) The correct-answer variety.

Where out of the options, only one is absolutely correct e.g. Which of the following is the largest town in Kenya? A) Mombasa B) Kisumu C) Nairobi D) Nakuru

b) The best-answer variety.

Consists of a stem followed by two or more suggested responses that are correct, appropriate in varying degrees, or down-right wrong (examinee responds with an opinion) e.g. Which of the following is the leading foreign exchange earner of Kenya?

A) Coffee B) Horticultural products C) Tourism D) Soda ash

c) The multiple-response variety.

Is where a number of clearly correct answers exist and the examinee is instructed to mark all the correct responses e.g. Which of the following are **not** capital cities in Africa. Mark the correct responses. A) Mogadishu B) Dar es Salaam C) Lagos D) Ouagadougou

d) The incomplete-statement variety.

Is where a portion of the stem is incomplete rather than a direct question e.g. The capital city of the Republic of South Africa is _____.

e) The negative variety.

It is where the examinee is to mark the response that does not correctly answer the question i.e. the least satisfactory answer e.g. Three of the following are major agricultural towns in Kenya. Which one is not? A) Bungoma B) Eldoret C) Kitale C) Kericho

f) The substitution variety

It is where samples of originally well written prose or poetry are systematically altered to include errors in punctuation, spelling, word usage and similar conventions. Selected words or phrases in these rewritten passages are underlined and identified by a number. Several possible substitutions for each critical phrase are provided and the examinee is asked to select the phrase (original or alternative) that provides the best expression e.g. Mr¹ Wangila has been the Principal² of WUCST³ since the inception of the college⁴.

(Professor, Doctor, Vice Chancellor, WUST, MMUST, Campus, University, University college)

g) The incomplete-alternatives variety

Is where incomplete or coded alternatives are used e.g. Which of the following is the fourth colour in the rainbow? A) Y B) G C) V D) G

h) The combined-response variety

Consists of an item stem followed by several responses, one more of which may be correct. The examinee is to choose the set of code letters or numerals which designate the correct responses. This variety tests a mastery of sets of facts and complex organization and comparative evaluation of facts or concepts e.g. Below are political parties in Kenya. (i) PNU (ii) ODM-K (iii) ODM (iv) GNU (v) KANU.

Which of the following combination has Kenya's past and current heads of state been associated with? A) (i) and (iii) B) (i) and (v) C) (iii) and (v) D) (ii) and (iv)



List several national examinations done in Kenya. For each of the listed examination, describe the types of test item used.

Suggestions for Constructing Multiple-Choice Items

- i. Select problems which present real problem to the examinees and call for critical thinking.
- ii. Select distracters which are attractive and plausible so that weak students can more often select them.
- iii. There should be only one key and no unintentional help/clue should be given.
- iv. The stem should be clear and responses should not borrow phrases from the stem.
- v. Avoid the use of negatives but if they must be used, they should be underlined, capitalized or italicized.
- vi. The key and the detractors should be more or less for equal length and should be short.
- vii. Avoid making the correct answer to the items appear in a fixed pattern.
- viii. Avoid the use of none of the above or all of the above. If not make them the correct detractor.



Look for past paper questions and make a list of errors made therein. Suggest how the question should have been set.

E. Maps, Diagrammatic and Pictorial Test Items

These are questions that require interpretation, recognition of parts or features etc. The following should be considered when designing such test items.

- i. Maps, pictures and diagrams must be simple and clear.
- ii. Do not shade pictures as they tend to be complicated beyond recognition.
- iii. Those with poor drawing skills should trace or use actual /real pictures, maps or diagrams.
- iv. Descriptive titles should be given to maps, pictures and diagrams and where necessary they should be framed.

? Draw the map of Kenya and construct at least five (5) questions based on the drawing?

CONSTRUCTION OF ESSAY QUESTIONS

When to use essay questions.

1. When the group is small and the test is not to be re-used.
 - Do not remind the candidates of the time left frequently. This can be done after 1hr or so or after completing one section of the paper.
 - Examination timetable should be released and given at least one a week in advance to enable students prepare adequately.

EXAMINATION CHEATING

It means to act dishonestly or unfairly in order to win an advantage or profit. It means to deceive and involves dishonest tricks in order to pass exams.

Methods used.

- Impersonation- sitting an exam on behalf of somebody.
- Gaining access to exam papers or confidential material or information related to the exam prior to sitting of the exam.
- Deliberate attempt to obtain or pass information concerning the exam when it is in progress. Information may be obtained from fellow students, invigilators, teachers or smuggled materials, whispering or “flashing” answers. Occasionally it involves seeking to go for a call of nature only to refer to information concealed somewhere.
- Practical subjects- teachers help in setting up equipments or offering answers or over scoring by teachers-in-charge.
- Use of mobile phones to text the answers to a candidate before or during the exam.
- Writing on the shirt sleeves, petticoats, desks or the thighs particularly by female university students.

Causes of Cheating

- Academic weakness of some of the students/teachers.
- Euphoria attached to exam results-goods grades are a source of pride to self, families and institutions.
- Need to excel due to stiff competition.
- Corruption and lack of transparency especially those charged with the responsibility of handling exam materials.
- Cheating as an easy way out. Quest for knowledge has seemingly lost meaning.
- Lack of commitment among students especially the lazy ones who don't take studies seriously.
- Congested curriculum and the belief that some subjects are difficult or impossible to pass.
- Uncertainty of employment among some course graduates leading to enrolment in others which may be demanding.

- Nature of examinations e.g. practicals. There is the temptation to look at one's neighbor's work.
- Traditional way of delivery lecturers with exams taking the same pattern. This makes it easy to guess and cheat.

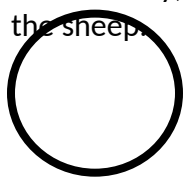
Effects of cheating

Diminishing credibility of examination as a measure of one's ability and in the examiner(s). Those who cheat can't compare in any way with those who don't.

- Loss of confidence in those charged with the handling of exams.
- Promotion to higher grade of education or training of the wrong people-who in turn perpetuate the practice.
- Kills teachers' morale especially those hard working ones.
- Kills morale of the hard working and honest students.
- Cause misunderstanding between the cheats and honest candidates especially when no action is taken against such.
- May often lead to result cancellation of the cheats with a doomed and painful future.
- Leads to repeating, suspension or expulsion from college causing more stress.
- Innocent students may suffer where results for a centre are cancelled.
- Compromises the education standards. Possible employers and other institutions doubt the authenticity of their academic credentials.
- Lead to criminal prosecution for the culprits and their accomplices and loss of job(s).

NB: Cheating in exams is just an aspect of moral decadence of the society. It is a manifestation of a sick society, devoid of a working culture and whose moral fiber has degenerated to irredeemable levels.

"Truly, truly, I say to you, he who does not enter the sheep fold by the door, but climbs in by another way, that man is a thief and a robber; but he who enters by the door is the shepherd of the sheep.



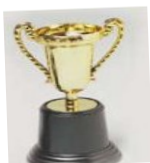
Learning Outcomes

You have finished topic 1. The learning outcomes are listed below. Place a (✓) in the column which reflects your understanding.

No.	Learning Outcome	Agree	Disagree
1	I can define the term		
2	I can explain the		
3	I can explain the		
4	I can discuss		

If for whatever reason you have put a tick on any of the statements, go back to the section before you proceed.

However, if you have ticked “agree” on all the statements, you can proceed to the subsequent section



Congratulations! You can continue to the next Topic

TOPIC 2

FREQUENCY DISTRIBUTIONS AND GRAPHIC PRESENTATIONS

Introduction



In this topic, you will learn more about common concepts used in statistics.. You will also get to know the various categories of Children in Need of Special Protection (CNSP) and the efforts the government is making to lessen their problems.

2.1 Objectives



By the end of the topic, you should be able to:

- Prepare a frequency table from given raw data.
- Draw a histogram and frequency curve/polygon from given data correctly.
- Draw and interpret a more-than and a less-than frequency curve (Ogive)
- Draw and interpret various shapes of frequency curves.

2.2 Topic 2 consists of the following sub-sections:

Section 1: Statistical Concepts in Tests and Measurement

Section 2: Frequency distributions and graphical presentation

Section 3: Stated and real class limits

Section 4: Histogram

Section 5: Frequency polygons and curves

Section 6: Skewness and kurtosis of a distribution

Now, let us deal with each of these sub-sections in detail.

2.2.1 STATISTICAL CONCEPTS IN TESTS AND MEASUREMENT

1. Statistics-the science of collecting data in a systematic manner, examining those data and making inferences from the data.

2. Statistic - a no that describes a characteristic of a sample e.g. 21.
3. Population – a complete set of individuals, objects, or measurement having some common characteristics.
4. Sample-a subject or part of population e.g. 3rd year B.Ed. female students.
5. Data – numbers or measurements that are collected as a result of observation. Interview etc e.g. PSY 311 CAT I scores.
6. Parameter-any characteristic of a population that is measurable e.g. Height/Weight.
Parameters are often inferred values based on sample statistics.
7. Variables-any characteristic of a person, group, or environment that can vary or denotes a difference e.g. IQ, height. There are two classes of variables:
 - a) Discontinuous variables/discrete variables: Are variables for which the values can only be whole numbers. There are no intermediate values between each number e.g. no of kids in a family.
 - b) Continuous variables: Are variables that can assume any value. There is an infinite no of values between any two numbers e.g. height, weight etc.
8. (i) Independent variable
The variable that can experiment use to describe or explain differences in the dependant variable or to cause change in the dependant variable.
- (ii) Dependent variable
It is an outcome of interest e.g. some aspect of behaviors that is observed and measured by a researcher in order to assess the effects of the independent variable.
9. Constant - a number that represents a construct that does not change e.g. $\pi = 3.1416$ or 1 ft =12 inches or the number of days in the month of January.



**The effect of large amount of money on a pupil's academic performance.
Which is the independent and dependent variables?**

Types of Statistics

1. Descriptive Statistics

Used to organize and summarize masses of numerical data e.g. frequency distributions, graphs, means, median, standard deviation, variance etc. Helps us discuss and understand data e.g. referendum.

2. Inferential statistics

It is also called inductive statistics or statistical inference. Is a collection of statistical techniques that allow one to make generalizations about population parameters based on sample statistics, to determine if there is a systematic relation between independent variable and the dependent variable, and to determine if there is a cause and effect relation between the independent variable and dependent variable e.g. Pearson product moment correlation coefficient.

Levels of measurement/scales

1) Nominal level/scales

It refers to data that can only be counted and put into categories. There is no particular order of the categories. Has the property of identification and nothing more e.g. serial number or name.. The number used in a nominal scale does not represent any quantity.

2) Ordinal scale

It is a basic form of quantitative measurement that indicates a numerical order such e.g. $2 < 3$ or $5 > 4$, i.e. the order and a succession of the numbers may be from top to bottom, greater to least, highest to lowest etc on some property. However, it lacks the elements of additively i.e. additions or subtractions are meaningless.

3) Interval scale

It is sometimes called equal internal scale. It is a measurement that has equal units of measurement and an arbitrary zero e.g. John is four inches taller/shorter than Peter. The difference in magnitude is based on some arbitrary starting point-the real heights of John and Peter remain unknown or 0°C does not mean that there is no temperature.

4) Ratio scale

Is a measurement that has equal units of measurement and an absolute zero point i.e. the zero point is real and indicates total absence of the property measured e.g. if you have zero shillings or there is zero weight means there is nothing at all. Or if Mary weighs 100kgs and Jane weighs 50kgs, it means Mary is twice as heavy as Jane.

Do you watch football? Which type of scale does number 10 assigned to a footballer represent?

Importance of studying statistics

- (1) In order to plan appropriate procedures, interpret and communicate findings in an intelligible manner.
- (2) Enables an individual consume research findings as published in various media e.g. newspapers, journals etc.
- (3) Enable educators interpret scores from class tests and major examinations correctly.

2.2.2 FREQUENCY DISTRIBUTIONS AND GRAPHICAL PRESENTATION

- Raw data can only be understood and interpreted when organized and summarized in some meaningful way. This is done using:
 - (a) Frequency
 - (b) Histograms
 - (c) Frequency polygons/curves
 - (d) Ogives
 - (e) Charts
 - (f) Line graphs etc.

FREQUENCY DISTRIBUTION

It is a grouping of data into categories showing the number of observations in each category

A. Frequency Distribution of Ungrouped Data

- It is done by arranging the scores in order, from the highest to the lowest or vice versa
- A tally (/) mark is made next to each score (attribute) whenever it occurs.
- The frequency – symbolized by the first letter of the word (lower case) summarizes the total number of tallies for each score.

Cumulative Frequency

- It refers to the number of scores in a frequency distribution that are within and below a specified frequency or class.

Example

Prepare a frequency distribution for the CAT scores in a Math class of 14 students.

4 2 6 7 4 4 6 7 9 5 4 3 5 5

Solution

X	Taly	<i>f</i>	<i>cf</i>
2	/	1	1
3	/	1	2
4	////	4	6
5	///	3	9
6	//	2	11
7	//	2	13
8	/	<u>1</u>	14

$$\Sigma f = 14$$

B. Frequency Distribution of Grouped Data

Grouping into class intervals involves “collapsing the scale” and assigning scores to mutually exclusive and exhaustive classes where the classes are defined in terms of the grouping intervals used.

Why group data?

- i. It is tedious and time wasting to deal with a large number of cases spread over many scores unless using a computer.
- ii. Some of the scores have very low frequency counts such that maintaining them as separate entities will not be justified.
- iii. Classes provide a concise and meaningful summary of the data.

Procedure for Establishing Class Intervals

Step 1: Find the difference between the highest and lowest score values contained in the original data. Add 1 to obtain the total number of scores or potential scores.

Step 2: Divide the figure by the number of class intervals that will provide the best summary of the data to obtain the number of scores or potential scores in each class intervals.

In most cases, 10-15 intervals will be adequate. If the resulting value is not a whole number (and it usually is not), round to the nearest odd number so that a whole number will be the mid –point of the class interval. However, this rule is not a must

Step 3: Add (W-1) to the minimum value of the lowest class to obtain the maximum score of the lowest class.

Step 4: The next higher class begins at the integer following the maximum score of the lower class.

Repeat step 3 to get the upper end of this class.

Step 5: Assign each obtained score to the class within which it is included.

Example

Below are ages of an ECD group of children. Prepare a frequency distribution.

2	5	8	9	3	5	7	1	8	10
10	3	6	11	14	8	6	12	4	7

Solution

Step 1: Lowest value = 1; Highest value = 14

$$\sqrt{(14 - 1) + 1} = 4$$

Step 2: Class width = $\frac{14}{4} = 3.5$, rounded off to 4

6

Step 3: $1 + (4-1) = 4$ class interval is 1 - 4

Step 4: $4 + (4-1) = 7$. Next class interval is 4 - 7 etc.

Class	Tally	<i>f</i>	<i>cf</i>
1 - 4	//	2	2
4 - 7	///	3	5
7 - 10	////	4	9
10 - 13	////	5	14
13 - 16	///	3	17
16 - 19	//	2	19
19 - 22	/	1	20

$$\sum f = 20$$



Are you a class teacher? Prepare a grouped frequency distribution of pupils in your class based on age.



SELF-TEST 2

Below are weights (in pounds) of 50 children in a refugee camp.

82	89	97	114	69	85	91	62
79	113	83	65	98	119	102	89
90	99	64	84	76	107	94	123
92	86	104	110	91	101	84	72
105	96	65	74	77	95	88	93

Prepare a frequency distribution for the data.

(5 mks)

2.2.3 STATED AND REAL CLASS LIMITS

- Continuous variables can take on an unlimited number of intermediate values. For this reason, numerical values of continuously distributed variables are always approximate.
- In a continuous distribution, each class interval has two class limits, the lower and upper limits.
- These class limits leave slight gaps between adjacent classes and are referred to as **Stated** or **Apparent** class limits.
- Stated/Apparent class limits mark boundaries of classes which do not overlap. They are normally expressed in whole numbers.
- **Real** or **True** class limits on the other hand specify the limits within which the true value falls.
- True/Real class limits are obtained by subtracting lower apparent/stated class limit and adding the same to the apparent/stated upper class limit.

Example

Apparent/stated Class Limits	Real/True Class Limits
5 - 9	4.5 - 9.5
10 -14	9.5 - 14.5
15 -19	14.5 - 19.5
20 - 24	19.5 - 24.5
25 - 29	24.5 - 29.5
30 - 34	29.5 - 34.5



When calculating certain statistics for grouped data, True/Real limits of the class interval(s) will be used.

Class midpoint

- The midpoint of a class, often called a class mark, is determined by going halfway between either the stated or true class limits.
- It is obtained by adding the lower and upper limits and dividing the total by two.

2.2.4 HISTOGRAM

- It is a form of bar graph used with interval or ration scaled frequency distributions.
- Each bar represents a single class. In behavioural/social sciences, the X-axis represents class intervals (independent variable) while the Y-axis represents frequency (Dependent variable).
- To construct a histogram, either the stated or the true limits or the midpoints are used.
- An appropriate scale should be selected in the ratio of 3:5 representing the X and Y axes respectively. This is obtained using the formula

$$\text{Highest frequency} - \text{Lowest frequency} = \frac{\text{ } X \text{ } }{\text{No. of classes}}$$

The product is rounded off to the nearest whole number (This forms the class interval for the Y-axis). A descriptive title for the histogram should be clearly stated to provide the heading.

Example

Class	f	True Class Limits
5 – 9	3	4.5 – 9.5
10 – 14	4	9.5 – 14.5
15 – 19	8	14.5 – 19.5
20 – 24	3	19.5 – 24.5
25 – 29	2	24.5 – 29.5

$$8 - 2 = \frac{6}{5} = 1.2 \simeq 1$$



SELF-TEST 3

Below are scores for a Standard seven class in a Science test.

Class	f
9-11	1
12-14	3
15-17	9
18-20	14
21-23	10
24-26	4

Using the data above, construct a histogram.

(5 marks)

2.2.5 FREQUENCY POLYGONS AND CURVES

- Both the frequency polygon and frequency curve have the same structure except that the frequency polygon is plotted and joined by straight lines while a frequency curve is plotted and joined by a smooth curve.

- To construct a frequency polygon/curve for grouped data, the class midpoints are used and are scaled on the X-axis while the class frequencies are on the Y-axis.
- The straight lines are extended to the X-axis one class below and one class above with zero frequencies to create a polygon (many sided figure). The figure should always have a title.



A frequency polygon can also be obtained by joining the mid-points of the tops of histogram bars.

Example

Construct a frequency curve for the data below.

Class	f	Class mid Point
1 – 3	2	2
4 – 6	5	5
7 – 9	8	8
10 – 12	3	11
13 – 15	2	14



SELF-TEST 4

Construct a frequency polygon for the following data.

(5 marks)

Class	f
5 – 9	3
10 – 14	4
15 – 19	8
20 – 24	3
25 – 29	2

2.2.6 SKEWNESS AND KURTOSIS

Skewness and kurtosis are terms that describe the shape and symmetry of a distribution of scores.

SKEWNESS: It refers to whether the distribution is symmetrical with respect to its dispersion from the mean. If on one side of the mean has extreme scores but the other does not, the distribution is said to be skewed.

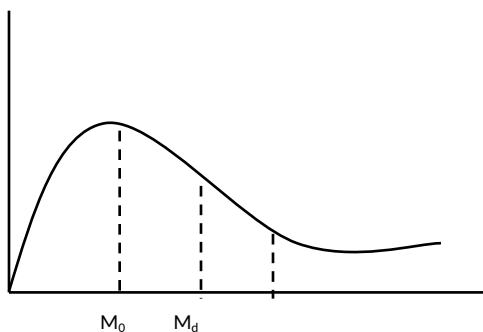
(a) Normal distribution

- If the dispersion of scores on either side of the mean are roughly symmetrical (i.e. one is a mirror reflection of the other) the distribution is said to be normal.



(b) Positively skewed distribution

- It is where the majority of scores are at the lower end of the distribution with a tail of scores at the upper end of the distribution.



In a positively skewed distribution, $\bar{X} > M_d > M_0$

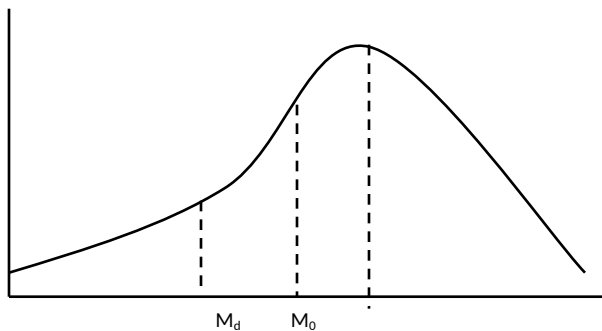


In a class test, it would mean that majority of the students scored below the class mean implying that;

- the test items may have been above the ability level of the students
- majority of the students are of below average ability
- the concept being tested may not have been well understood by the students

(c) Negatively skewed distribution

It refers to a distribution where the scores are clustered at the upper end of the scale with a tail of scores at the lower end of the distribution.



In a negatively skewed distribution, $\bar{X} < M_d < M_0$



In a class test, it would mean that majority of the students scored above the class mean implying that;

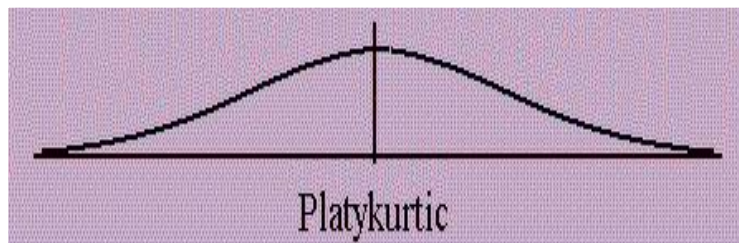
- majority of the students may be of above average ability
- the test items may have been easy
- the concept being tested may have been well understood by the students

? As a teacher, if you gave your class a test and the number of students who scored above the class mean is the same as those who scored below the class mean, what interpretation would you make?

KURTOSIS: It refers to the weight of the tails of a distribution. Distributions where a large proportion of the scores are towards the extremes are said to be platykurtic. If, on the other hand, the scores are bunched up near the mean, the distribution is said to be leptokurtic. A normally distributed distribution of scores is said to be mesokurtic.

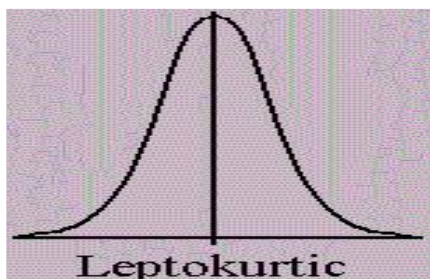
i. Platykurtic distribution

It is where the scores are spread across forming a “platform-like” distribution.



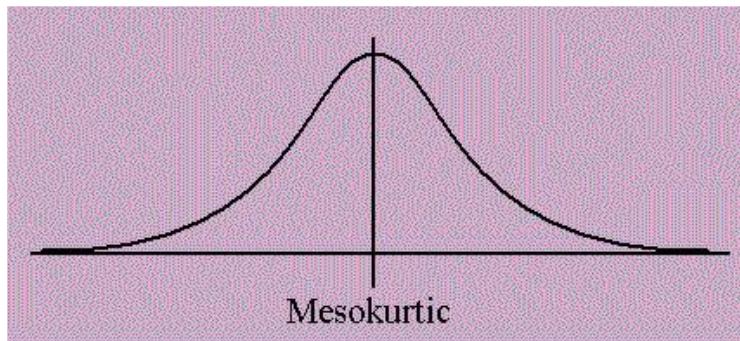
ii. Leptokurtic distribution

It is characterized by the piling up of scores in the centre of the distribution.



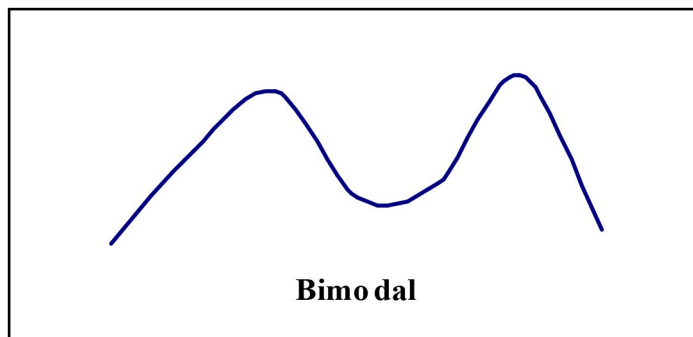
iii. Mesokurtic

It refers to a normally distributed set of data.



iv. Bimodal distribution

It is where a variable has a high concentration of frequencies around two separate values or where frequency distributions of two different populations are represented in single graph e.g. average adult height of males and females.



CUMULATIVE FREQUENCY POLYGON/CURVE OR OGIVE

- It is used to determine the number of observations that lie above or below certain values. There are of two types namely a less than and a more than cumulative frequency polygons.

1. A less than cumulative frequency polygon/curve

- It tells how many items in the distribution have a value less than the upper class limit of the first class, less than the upper limit of the second class etc.
- It is used for estimation purposes. It answers questions such as “How many values are less than 40?” or “What percent of the values are less than 25?”

- ✦ To construct a less than cumulative frequency polygon, the upper true class limits and cumulative frequencies, are plotted. They are joined with a smooth curve.

2. A more than cumulative frequency polygon/curve

- ✦ It tells how many items in the distribution have a value greater than or equal to the value of the lower limit of the first class, greater than or equal to the value of the lower limit of the second class etc.
- ✦ It answers questions such as “How many scores in the distribution are more than ____?” or what percent of the scores are more than ____?”
- ✦ To construct a more than cumulative frequency curve the lower true class limits and cumulative frequency above (*cf*) are used.

Example

Class	<i>f</i>	<i>cf</i>	True class limits
6-8	2	2	5.5 - 8.5
9-11	3	5	8.5 - 11.5
12-14	4	9	11.5 - 14.5
15-17	7	16	14.5 - 17.5
18-20	13	29	17.5 - 20.5
21-23	4	33	20.5 - 23.5
24 – 26	2	35	23.5 - 26.5

Solution

$$35-2 = \underline{33} = 4.7 \simeq 5$$



SELF-TEST 5

The data below represents the weight in pounds of pupils in a public secondary school in Kenya. Draw a Less than cumulative frequency polygon to depict the data.

Class	f
109-119	1
119-129	4
129-139	17
139-149	28
149-159	25
159-169	18
169-179	13
179-189	6
189-199	5
199-209	2
209-219	<u>1</u>
$\Sigma f = 120$	



SELF-TEST 5

Q. Below are scores in an Educational Psychology test.

60 33 52 65 47 65 57 74 66 46 73 42
43 64 55 22 63 45 74 57 45 70 64 58
50 25 35 34 27 38 51 29 33 41 35 50
41 61 55 73 59 53 45 57 41 78 55 48
54 47 68 54 60 76 64 39 64 53 65 35

Using $i = 10$ and starting with 20-29,

- a) Prepare a grouped frequency distribution table for the data. (3 mks)
- b) Draw a frequency curve to depict the data (3 mks)
- c) What type of distribution do the scores form? (2 mks)
- d) Draw a less-than frequency curve. How many values are less than -----? What percent of the values are less than?" (7 mks)

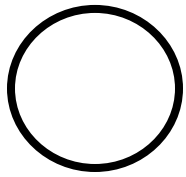


Summary

In this topic we have learnt various concepts commonly used in statistical applications. But more importantly, we have learnt how to prepare a frequency distribution from raw data and how to represent the data using various graphical representations such as histograms, frequency polygons and ogives. We also learnt about the various shapes produced by different sets of data and what such shapes mean to the classroom teacher.

Score Board

Score	Comment	Remarks
0-6	Poor	Go back and read through the whole topic
7-9	Satisfactory	Go back and read the sections that are not clear
10-12	Good	You can proceed but after looking at the questions again
13-15	Excellent	Proceed to the next topic



Learning Outcomes

You have finished topic 2. The learning outcomes are listed below. Place a (✓) in the column which reflects your understanding.

No.	Learning Outcome	Agree	Disagree
1	I can explain the various concepts commonly used in statistical applications		
2	I can tabulate and depict in a histogram, frequency curve/polygon and ogive a given set of data.		
3			
4	I can differentiate with examples the skewness and kurtosis of a distribution		

How many of these statements have you responded with “Disagree”? If for whatever reason you have done so, go back to the section before you proceed.

However, if you have ticked “agree” on all the statements, you can proceed to the subsequent section



Congratulations! You can continue to the next Topic

TOPIC 3

3.0 MEASURES OF CENTRAL TENDENCY

Introduction



In this topic, you will learn more about measures of central tendency. These refer to descriptive statistics that indicate the central location of a distribution of observations such as the mode, median and mean. You will also get to know when these measures can be used and their advantages and disadvantages.

3.1 Objectives



By the end of the topic, you should be able to:

- 👤 Compute the mode of ungrouped and grouped data.
- 👤 Explain when the mode can be used and state its advantages and disadvantages.
- 👤 Explain when the median can be used and state its advantages and disadvantages.
- 👤 Explain when the mean can be used and state its advantages and disadvantages

2.2 Topic 3 consists of the following sub-sections:

Section 1: The mode

Section 2: The median

Section 3: The mean

Section 4: Mean, mode and median compared

Now, let us deal with each of these sub-sections in detail.

3.1 THE MODE

Mode for Ungrouped Data

- It is the value in a distribution with the highest frequency i.e. the most recurring value. Where the mode does not exist, it is usually estimated e.g.

- i) No mode exists in a distribution where values have the same frequency e.g.

1 3 4 5 8 9

Where one score has higher frequency than others in a distribution, the score is the mode e.g. 1 3 4 4 5 8 9

Mode is 4

- ii) Where two adjacent scores have the same frequency and this frequency is the highest in the distribution, the mode is the average of the two modes e.g.

1 3 4 4 5 8 8 9

$$\text{Mode} = 4 + 8 = \frac{12}{2} = \underline{6}$$

- iii) Where the modes are not adjacent, we shall have multiple models. Such modes are reported without averaging e.g. as in a bimodal distribution e.g.

1 3 4 4 5 8 9 9

The modes are 4 and 9

Mode for Grouped Data

There are two methods of estimating the mode

- (a) Using the interpolation formula
- (b) Using the graphic representation

(a) Interpolation Formula

Step I: Determine the modal class (class with the highest frequency)

Step II: Calculate D_1 = Difference between the largest frequency and the frequency immediately preceding it.

Step III: Calculate D_2 = Difference between the largest frequency and the frequency immediately following it.

Step IV: Use the interpolation formula below

$$\text{Mode } (M_0) = L + \frac{D_1}{D_1 + D_2} i$$

Where L = True lower limit of the modal class

D_1 = Difference between the f of the modal class and the f of the class immediately preceding

D_2 = Difference between the f of the modal class and the f of the class immediately following it in the distribution

i = the class width/interval

Example

Compute the mode of the following frequency distribution.

Class	f
20-24	8
25-29	14
30-34	12
35-39	7
40-44	3

Solution

$$\begin{aligned}M_o &= L + \frac{\frac{D_1}{f_1 + f_2}}{\frac{D_1}{f_1 + f_2} + \frac{D_2}{f_2 + f_3}} \times i \\&= 34.5 + \frac{\frac{6}{86 + 20}}{\frac{6}{86 + 20} + \frac{5}{20 + 10}} \times 5 \\&= 24.5 + (0.75 \times 5) \\&= 24.5 + 3.75 \\M_o &= 28.25\end{aligned}$$

(b) Graphic Representation Method

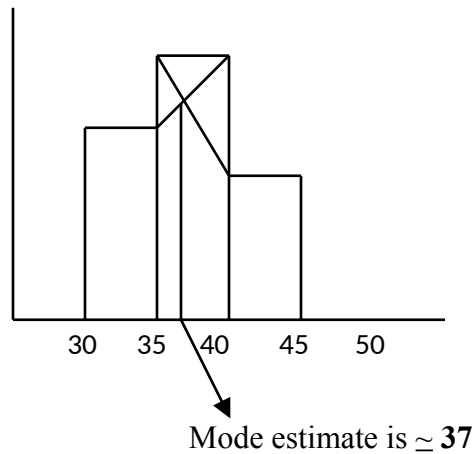
- Construct three histogram bars, representing the class with the highest frequency and the ones on either side of it.
- Draw two lines from the highest ends of the modal class to the point where the preceding and following class levels meet.
- The mode estimate is the X- value corresponding to the intersection of the lines.

Example

Using the graphic method, find the mode of the following data.

Class	<i>f</i>
20 – 25	2
25 – 30	4
30 – 35	5
35 – 40	7
40 – 45	3
45 – 50	1

Solution



When the mode can be used

- i. When the average is easily and quickly desired
- ii. When the data are non-quantitative e.g. mode of dressing (most popular).
- iii. When the singularity or multiplicity of group formation is to be investigated.

? What is the mode (colour) of school uniforms in your locality?

Advantages

1. It can be obtained for any set of data.
2. It is easy to understand.
3. It is not affected by extreme values.
4. It can be obtained for quantitative data.

Disadvantages

1. Not all sets of data have a modal value.
2. Some sets of data have multiple modal values hence are difficult to interpret.
3. The mode lacks useful mathematical properties i.e. it cannot be used for further calculations.



SELF-TEST

Class	<i>f</i>
20 - 29	4
30 - 39	8
40 - 49	12
50 - 59	16
60 - 69	13
70 - 79	7

- Compute the mode of the data below using the interpolation formula.
- Using the graphic representation method, find the mode.

3.2 THE MEDIAN

It is the point in a distribution that has equal number of scores above and below it. It is the mid point of a distribution; the value at the 50th percentile.

CALCULATING THE MEDIAN

Below are statistics for a number of car accidents in eleven (11) months in busy town.

16 11 12 10 13 17 12 14 12 14 15

Step I: Arrange the numbers from the lowest to the highest or vice versa

10 11 12 12 12 13 14 14 15 16 17

Step II: Add 1+N and divide the total by 2, i.e.

$$M_d = \frac{N+1}{2} = \frac{11+1}{2} = \frac{12}{2} = 6$$

Step III: Starting with the lowest value, count up to the sixth value. The sixth value is the median.

10 11 12 12 12 **[13]** 14 14 15 16 17

↓
Median



If there is an even number of values (scores), the median is half way between the two middle value e.g.

12 13 14 15 16 17

$$\frac{N+1}{2} = \frac{6+1}{2} = \frac{7}{2} = 3.5$$

To obtain the median, the two adjacent values are added and divided by 2, i.e. $\frac{14 + 15}{2} = 14.5$

MEDIAN OF A FREQUENCY DISTRIBUTION

Example

Find the median of the following frequency distribution of 30 scores in a statistics test

X	<i>f</i>	<i>cf</i>
11	1	1
14	2	3
15	7	10
17	14	24
19	4	28
20	<u>2</u>	30

$$\sum f = 30$$

Procedure

Step I: Divide $N+1$ by 2 to find the location of the middle frequency i.e.

$$\frac{N+1}{2} = \frac{30+1}{2} = \frac{31}{2} = 15.5$$

The 15.5th position lies within the 24th *cf*.

Step II: The median is identified by selecting the observation that corresponds to that value 17 (a satisfactory estimate of the median).

THE MEDIAN FOR GROUPED FREQUENCY DISTRIBUTION

Estimation of the median can be done using the interpolation formula.

Procedure

Step I: Form a cumulative frequency (cf) column

Step II: Find the value of $N/2$ (where $N = \sum f$)

Step III: Find the value of the cumulative frequency below the median class (cf_b)

Step IV: Find the frequency value within the median class (fw)

Step V: Calculate the median using the formula below;

$$\text{Median (Md)} = L + \frac{\frac{N}{2} - cf_b}{fw} \times i$$

Where L = true lower limit of the median class

N = Sample total

cf_b = Cumulative frequency up to the lower limit of the median class

fw = Frequency of the median class

i = Width of class interval

Example

Class	f	cf
20-24	2	2
25-29	14	16
30-34	29	45
35-39	43	88
40-44	33	121
45-49	9	130
	$\sum f = 130$	

$$L = 34.5$$

$$N = 130$$

$$Cf_b = 45$$

$$f_w = 43$$

$$i = 5$$

$$M_d = L + \frac{\frac{N}{2} - cf_b}{f_w} \times i$$

$$= 35 + \frac{\frac{130}{2} - 45}{43} \times 5$$

$$= 34.5 + (0.465 \times 5)$$

$$= 34.5 + 2.33$$

$$M_d = 36.83 \text{ (2 decimal places)}$$

When to use the median

1. When the distribution has extreme values (in a skewed data).
2. When the values at the end of a distribution are not known.
3. When the exact midpoint of the distribution is wanted i.e. the 50% point

Advantages

1. The concept is easy to understand and interpret.
2. It can be determined for any data set.
3. It is not easily affected by extreme values in a data set.

Disadvantages

1. The data must first be arranged in an array (ascending or descending order).
2. It lacks the useful mathematical properties i.e. it cannot be used for further computation.



SELF-TEST 9

The following data was obtained in an IQ test from a group of disadvantaged children in a slum area. Compute the median.

Class	<i>f</i>
75-79	3
80-84	4
85-89	18
90-94	20
95-99	10
100-104	8
105-109	5
110-114	2

3.3 THE MEAN

There are four types of means;

1. Arithmetic mean
2. Geometric mean
3. Harmonic mean
4. Quadratic mean



For the purpose of this course, only the Arithmetic mean will be looked at in detail. This is because it is what the classroom teacher uses in his/her daily teaching/learning activities.

Arithmetic mean

It is commonly referred to as the “average”. It is defined as “the sum of the values divided by the number of values” i.e.

$$\bar{X} = \frac{\sum x_i}{N}$$

Find the mean of 12 8 25 26 10

$$\bar{X} = \frac{12 + 8 + 25 + 26 + 10}{5} = \frac{81}{5}$$

$$\bar{X} = 16.2$$

The mean of a simple frequency distribution

Large data set is normally arranged into a frequency distribution. The above formula is not appropriate since it does not take account of the frequencies. The formula below is used.

$$\bar{X} = \frac{\sum x}{N}$$

Example

x	f	fx
10	2	10 x 2 = 20
12	8	12 x 8 = 96
13	17	13 x 17 = 221
14	5	14 x 5 = 70
16	1	16 x 1 = 16
19	<u>1</u>	<u>19 x 1 = 19</u>
	$\sum f = 34$	$\sum fx = 442$

$$\bar{X} = \frac{\sum fx}{N}$$

$$= \frac{442}{34}$$

$$\bar{X} = 13$$

THE MEAN OF GROUPED FREQUENCY DISTRIBUTION

Procedure

Step I: Find the group (class) midpoints (x) as representative x-values

Step II: Estimate the totals of the values in each group using fx i.e. fx

Step III: Add the totals to form an estimate of the total of all values i.e. $\sum fx$

Step IV: Divide $\sum fx$ by the total number of items i.e. $\frac{\sum fx}{\sum f}$.

$$\bar{X} = \frac{\sum fx}{\sum f}$$

Example

Class	f	Midpoint (x)	fx
0-4	2	2	4
5-9	4	7	28
10-14	12	12	144
15-19	19	17	323
20-24	14	22	308
25-29	7	27	189
30-34	<u>2</u>	32	<u>64</u>
	$\sum f = 60$		$\sum fx = 1060$

$$\begin{aligned}\bar{X} &= \frac{\sum fx}{\sum f} \\ &= \frac{1060}{60}\end{aligned}$$

$$\bar{X} = 17.67$$



SELF-TEST

Find the mean for the following data set

<i>Age (yrs)</i>	<i>f</i>
20-25	2
25-30	14
30-35	29
35-40	43
40-45	33
45-50	9

CALCULATION OF THE MEAN BY THE “ASSUMED MEAN”

The formula used is as below

$$\bar{X} = \frac{\sum f(x - A)}{\sum f}$$

Where: A = assumed mean i.e. the midpoint of some class

$\sum f(x - A)$ = is the product of f and deviation scores $(x - A)$

$\sum f$ = the number of observations.

Example

Taking 17 as your assumed mean, find the true mean for the following distribution.

Class	f	x	$x-A$	$f(x-A)$
0-4	2	2	-15	-30
5-9	4	7	-10	-40
10-14	12	12	-5	-60 (Sub total =130)
15-19	19	17	0	0
20-24	14	22	5	70
25-29	7	27	10	70
30-34	<u>2</u>	32	15	<u>30</u> (Sub total = 170)
	$\Sigma f=60$			$\Sigma f(x-A) =40$

$$\begin{aligned}\bar{X} &= 17 + \frac{40}{60} \\ &= 17 + 0.67 \\ \bar{X} &= 17.67\end{aligned}$$

When to use the Mean

1. When the values (scores) are distributed symmetrically around a central point i.e. when the distribution is not badly skewed.
2. When the measure of central tendency having the greatest stability is wanted.
3. When other statistics (e.g. SD, coefficient of correlation etc) are to be computed later.

Advantages

1. It uses all values in the distribution hence its more stable.
2. It is used to draw inferences (conclusions)

Disadvantages

1. It is unduly affected by extreme values.
2. It is difficult to compute compared to the mode and median.



SELF-TEST

Taking 42.5 as your assumed mean, find the true mean for the following data set.

<i>Age (yrs)</i>	<i>f</i>
20-25	2
25-30	14
30-35	29
35-40	43
40-45	33
45-50	9

3.4 MEAN, MODE AND MEDIAN COMPARED

Interpretation

Example

In one of the previous examples above, the following mean, median and mode were obtained.

$$\bar{X} = 17.67$$

$$M_d = 17.65$$

$$M_o = 17.4$$

If: $\bar{X} > M_d > M_o$ i.e. Positive Skew

$\bar{X} < M_d < M_o$ i.e. Negative Skew

$\bar{X} = M_d = M_o$ i.e. Normal Distribution

Thus, in the above example, $\bar{X} > M_d > M_o$, hence the distribution is positively skewed. Most of the scores lie below the mean.

Pearson Measures of Skewness

$$Psk = \frac{Mean - Mode}{SD} = \frac{\bar{X} - M_o}{SD}$$

Assume as in the example above the $\bar{X} = 17.67$

$$M_o = 17.4$$

$$Sd = 6.60$$

$$Psk = \frac{17.67 - 17.4}{6.60}$$

$$= \frac{0.27}{6.60}$$

$$Psk = 0.04$$

Interpretation

$Psk < 0$ = Negative Skew

$Psk > 0$ = Positive Skew

$Psk = 0$ = Normal distribution

In this example, $Psk = 0.04 > 0$. Thus the distribution is positively skewed implying that most values/scores lie below the mean.



Why is the school mean in KCPE sometimes deceptive to the layman?



SELF-TEST

Below are scores of 80 students in an Educational Planning and Management test.

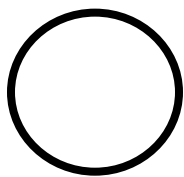
23 84 61 87 43 72 62 78 69 47
81 94 59 76 33 29 57 49 51 69
58 81 58 43 76 43 64 55 22 63
55 67 75 40 73 92 65 82 50 86
75 65 72 53 65 80 57 73 36 33
61 62 84 46 77 55 74 53 70 69
70 62 61 73 72 85 50 86 45 30
30 34 28 41 43 35 36 37 32 36

Using $i = 10$ and starting with 20-29,

- Prepare a grouped frequency distribution for the data. (3 marks)
- Using the graphic representation method, find the mode (2½ marks)
- Compute the mean and median. (5 marks)
- Using a suitable technique, determine the skew. (3½ marks)
- Comment on (d) above. (1 mark)

Score Board

Score	Comment	Remarks
0-6	Poor	Go back and read through the whole topic
7-9	Satisfactory	Go back and read the sections that are not clear
10-12	Good	You can proceed but after looking at the questions again
13-15	Excellent	Proceed to the next topic



Learning Outcomes

You have finished topic 3. The learning outcomes are listed below. Place a (√) in the column which reflects your understanding.

No.	Learning Outcome	Agree	Disagree
1	I can define the		
2	I can explain the		

3	I can explain the		
4	I can discuss		

If for whatever reason you have put a tick on any of the statements, go back to the section before you proceed.

However, if you have ticked “agree” on all the statements, you can proceed to the subsequent section



Congratulations! You can continue to the next Topic

TOPIC 4

MEASURES OF DISPERSION/VARIABILITY

Topic 4 has the following sections:

Section 1: Range

Section 2: Variance**Section 3: Standard deviation****Section 4: Interquartile range/deviation****Section 5: Percentiles****Meaning**

Measures of dispersion or variability describe how scattered a distribution of values/scores is. They show the degree to which individual scores differ from one another in a data set. Such measures include;

- i) The range
- ii) The variance
- iii) The standard deviation
- iv) The interquartile range/quartile deviation
- v) Percentiles.

THE RANGE

It refers to the difference between the highest and lowest values in a set of data.

$$\text{Range} = \text{Highest value} - \text{Lowest value}$$

Example

Find the range of the following data.

13 17 14 16 10 9 11

$$\text{Range} = 17 - 9 = 8$$

When to use the Range

1. When the data are too scant or too scattered to justifying the computation of a more precise measure of variability.
2. When knowledge of extreme scores or a total spread is all that is needed.

Advantage

- a) It is easy to determine and understand.

Disadvantages

- a) It only takes two values into account and is therefore affected by extreme scores.
b) It is unreliable when N is small or when there are large gaps in the frequency distribution.

THE VARIANCE

- It is the average of the squared differences between the mean and the observed scores.
- It is denoted by the symbol s^2 or v or σ^2 .

Calculating Variance for Ungrouped Data

There are two formulas.

- a) Definitional formula- used when each score and the mean are not whole numbers.
b) Computational formula- ideal where the sample is large and the scores and mean are whole numbers.

There are two commonly used formulae, the definitional and computational formulae.

Definition formula

$$S^2 = \frac{\sum d^2}{N} \quad \text{or} \quad S^2 = \frac{\sum (X - \bar{X})^2}{N}$$

Computational formula

$$S^2 = \frac{\sum x^2}{N} - \frac{(\sum X)^2}{N}$$

Example

Find the variance for the following data.

9 11 7 8 10

Solution

Definition formula

X	$X - \bar{X}(d)$	$(X - \bar{X})^2(d^2)$
7	-2	4
8	-1	1
9	0	0
10	1	1
11	2	4
$\Sigma X = 45$		$\Sigma (X - \bar{X})^2 = 28$
$N = 9$		

$$S^2 = \frac{\Sigma (X - \bar{X})^2}{N}$$

$$= \frac{28}{9}$$

$$S^2 = 3.11$$

Computational formula

X	X²
7	49
8	64
9	81
10	100
11	121
$\Sigma X = 45$	$\Sigma X^2 = 415$

$$S^2 = \frac{\Sigma X^2}{N} - \bar{X}^2$$

$$= \frac{757}{9} - \frac{81}{9}$$

$$= 84.1 - 81$$

$$S^2 = 3.11$$



SELF-TEST

Calculate the variance of the following data set

10.4 14.7 13.6 14.4 16.1 18.5

Calculating the Variance for Grouped Data

$$S^2 = \frac{\sum f(X - \bar{X})^2}{N}$$

Example

Calculate the variance for the following set of data.

X	f	x	fx	$X - \bar{X}$	$X - \bar{X}^2$	$f(X - \bar{X})^2$
2 – 4	2	3	6	-5.82	33.87	67.74
5 – 7	4	6	24	-2.82	7.95	31.80
8 – 10	6	9	54	0.18	0.03	0.19
11 – 13	3	12	36	3.18	10.11	30.33
14 – 16	2	15	30	6.18	38.19	76.38
$\sum X = 17$		$\sum fx = 150$		$\sum f(X - \bar{X})^2 = 206.44$		

$$\bar{X} = \frac{\sum fx}{\sum f} = \frac{150}{17}$$

$$= 8.82$$

$$S^2 = \frac{\sum f(X - \bar{X})^2}{N}$$

$$= \frac{206.44}{17}$$

$$S^2 = 12.14$$



SELF-TEST

Below are scores in a History test. Calculate the variance.

Class	<i>f</i>
35-39	3
40-44	3
45-49	5
50-54	8
55-59	7
60-64	3
65-69	2

THE STANDARD DEVIATION

The standard deviation (SD) is the most stable index of variability. It is represented by the symbol *s* or *S* (sigma). The SD of a set of data is the square root of the variance.

$$s = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

Example

Calculate the standard deviation for the data below.

5 2 7 4 8

Solution

X	$X - \bar{X}$	$(X - \bar{X})^2$
2	-3.6	12.96
4	-1.6	2.56
5	-0.6	0.36
7	1.4	1.96
8	2.4	5.76
$\sum X = 28$		$\sum (X - \bar{X})^2 = 23.6$
$\bar{X} = \frac{\sum x_i}{N}$		

$$\bar{X} = \frac{28}{5} = 5.6$$

$$= \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

$$= \sqrt{\frac{23.6}{5}}$$

$$= \sqrt{4.72}$$

$$SD = 2.17$$



SELF-TEST

Compute the standard deviation for the following data.

9 7 10 9 11 8 9

STANDARD DEVIATION OF GROUPED DATA

- Step I:** Find the class midpoint as representative x-values.
- Step II:** Estimate the totals of the values in each group using fx i.e. fx
- Step III:** Add the totals to form an estimate of the total of all values i.e. $\sum fx$
- Step IV:** Square the x-values and multiply by their respective f to get $f(x^2)$. Add $f(x^2)$ i.e. $\sum f(x^2)$
- Step V:** Compute the standard deviation using formula below:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n} = \frac{\sum f x^2}{\sum f} - \left(\frac{\sum f x}{\sum f} \right)^2}$$

Example

Calculate the standard deviation for the following data.

X	f	x	fx	x ²	f(x ²)
2 – 4	2	3	6	9	18
5 – 7	4	6	24	36	144
8 – 10	6	9	54	81	486
11 – 13	3	12	36	144	432
14 – 16	2	15	30	225	450
	<u>$\sum X = 17$</u>		<u>$\sum fx = 150$</u>		<u>$\sum f(x^2) = 1530$</u>

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n} = \frac{\sum f x^2}{\sum f} - \left(\frac{\sum f x}{\sum f} \right)^2}$$

$$= \sqrt{\frac{1530}{17} - \left(\frac{150}{17} \right)^2}$$

$$= \sqrt{90 - 77.85}$$

$$= \sqrt{12.15}$$

$$= 3.49$$

Interpretation

The bigger the S , the larger the spread while the smaller the SD , the smaller the spread.

When to use the SD

1. When the statistic having the greatest stability is sought.
2. When extreme deviation should exercise a proportionally greater effect upon the variability.
3. When coefficient of correlation and other statistics are to be computed later.

Pearson Measure of Skewness

$$PSK = \frac{Mean - Mode}{SD} = \frac{\bar{X} - M_o}{s}$$

Assume the $\bar{X} = 36.54$

$$M_o = 37.92$$

$$s = 5.73$$

$$= \frac{36.54 - 37.92}{5.73}$$

$$= \frac{-1.38}{5.73}$$

$$= -0.24$$



If, $PSK < 0$ = Negatively skewed distribution

$PSK = 0$ = Normal distribution

$PSK > 0$ = Positively skewed distribution

In this example, $PSK < 0$. Therefore, the distribution is negatively skewed.



SELF-TEST

The data below was obtained from a group of 4th Year students in an EPM test.

Class	f
34-38	3
39-43	9
44-48	17
49-53	23
54-58	15
59-63	8
64-68	5

- Compute the mean, mode and standard deviation for the data set. (6½ marks)
- Using an appropriate technique determine the skew. (2 marks)
- Interpret your findings in (ii) above. (1½ marks)

QUARTILES

✦ A (size ordered) set of data can be split into four equal parts. The median divides the total set of data into two equal parts.

✦ When the lower half is divided into two equal parts, the value of the dividing variate is called the **lower quartile** or the **1st quartile**, denoted by Q_1 i.e. the point below which lie 25% of the scores.

✦ The values of the variate dividing the upper half is called the **upper quartile** or **3rd quartile** denoted by Q_3 i.e. the point below which lie 75% of the scores.

✦ The median is sometimes referred to as the **2nd quartile**, Q_2 e.g.

17 13 15 14 13 19 18

Size ordered, 13 13 14 15 17 18 19

Diagram illustrating the size-ordered data set: 13 13 14 15 17 18 19. Arrows point to the 2nd value (13), the 4th value (15), and the 6th value (18), representing the 1st, 2nd (median), and 3rd quartiles respectively.

Lower (1st) quartile

Middle (2nd) quartile
(Median)

Upper (3rd) quartile

Identification of the quartiles for a set

For an ordered set of data, the following method is used.

Q1 is the value of the $\frac{n+1}{4}$ th item.

Q3 is the value of the $\frac{3(n+1)}{4}$ th item.



Although the median is the middle quartile, the term “quartile” is often used to describe only the lower and upper quartiles, Q₁ and Q₃ respectively.

Example

Calculate the quartile for the following set of data.

11 16 15 18 14 19 17

Size ordering: 11 14 15 16 17 18 19

Q₁ is the value of the $\frac{7+1}{4}$ th = 2nd item, which is 14

Q₃ is the value of the $\frac{3(7+1)}{4}$ th = 6th item which is 18

The Quartile Deviation/Semi interquartile range (SIQR)

¶ The quartile deviation is defined as half the range of the middle 50% of items (i.e. the difference between the lower and upper quartiles divided by two).

¶ The formula used is;

$$q^d/SIQR = \frac{Q_3 - Q_1}{2}$$

Example

Calculate the quartile deviation for the following data set

11 16 15 18 14 19 17

$Q1 = 14$

$Q3 = 18$

$$q^d (SIQR) = \frac{Q3 - Q1}{2}$$

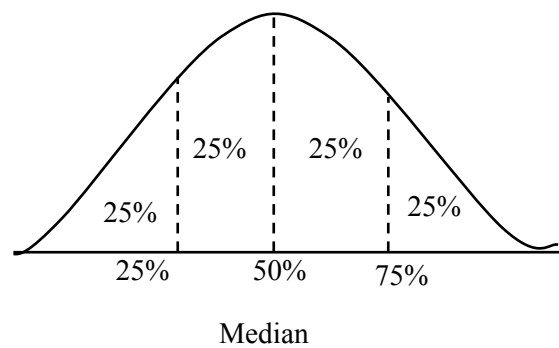
$$= \frac{18 - 14}{2}$$

$$= \frac{4}{2}$$

$$SIQR = 2$$

IDENTIFYING THE QUARTILE OF A FREQUENCY DISTRIBUTION

The quartiles split a distribution into four equal portions, which means that the area under the frequency curve is divided into four equal parts.



The Quartile Deviation for a Simple Frequency Distribution

Example

Calculate the median and quartile deviation for the following distribution.

X	<i>f</i>	<i>cf</i>	
4	4	4	$\frac{64+1}{4} = \frac{65}{4} = 16.25$
5	8	12	
6	10	22	$Q_1 = \frac{N+1}{4} \text{ th} = \frac{66}{4} \text{ th} = 16.25 \text{ item}$
7	11	33	
8	15	48	
9	10	58	$Q_1 = 6$
10	4	62	
11	2	64	$Q_3 = 3\left(\frac{N+1}{4}\right) = 3\left(\frac{65}{4}\right) = 48.75$

$$\begin{aligned} \text{qd/SIQR} &= \frac{Q_3 - Q_1}{2} \\ &= \frac{8 - 6}{2} \end{aligned}$$

$$\text{SIQR} = 1$$

$$Q_3 = 8$$

$$\text{Median} = \frac{N+1}{2} = \frac{65+1}{2} = 33$$

$$\text{Md} = 7$$



SELF-TEST

The scores below were obtained in Psychology test among 2nd Year School based students in MMUST.

X	<i>f</i>
14	8
16	10
17	16
18	21

20	14
22	11
23	7
24	3

Calculate the median and quartile deviation.

Calculation of the Quartiles of Grouped Data

For grouped data, the formula below is used.

$$Q = \frac{L + \left(\frac{N}{4} - Cumf \right) i}{f_q}$$

Where; L = the exact lower limit of the interval in which the quartile falls.

Cumf = Cumulative frequency up to the interval containing Q_i

f_q = the *f* of the interval containing the *q*

i = the class interval

Example

Calculate the quartile for the following distribution of scores in a Biology test.

Class	<i>f</i>	<i>cf</i>
5 – 9	3	3
10 – 14	5	8
15 – 19	9	17
20 – 24	7	24
25 – 29	4	28
30 – 34	<u>2</u>	30
N = 30		

$$\begin{aligned}
 Md &= \frac{L + \left(\frac{N}{2} - C f_b\right) i}{f_w} \\
 &= 14.5 + \frac{25 - 8}{9} \times 5 \\
 &= 14.5 + 0.78 \times 5 \\
 &= 14.5 + 3.89 \\
 Md &= 18.39
 \end{aligned}$$

$$\begin{aligned}
 Q_1 &= L + \frac{\left(\frac{N}{4} - Cumf\right) i}{f_q} \\
 &= 9.5 + \frac{\left(\frac{30}{4} - 3\right) 5}{5} \\
 &= 9.5 + 0.9 \times 5 \\
 &= 9.5 + 4.5 \\
 &= 14
 \end{aligned}$$

$$\begin{aligned}
 &= 19.5 + 0.79 \times 5 \\
 &= 19.5 + 3.93 \\
 &= 23.43
 \end{aligned}$$

$$\begin{aligned}
 \text{Therefore, } qd/SIQR &= \frac{Q_3 - Q_1}{2} \\
 &= \frac{23.43 - 14}{2} \\
 &= \frac{9.43}{2} \\
 &= 4.72
 \end{aligned}$$

Interpretation

- † The quartiles for Q_3 and Q_1 mark off the limits of the middle 50% of scores in the distribution.
- † The distance between these two points is called the interquartile range.
- † Q is $\frac{1}{2}$ the range of the middle 50% or the semi-interquartile range (SIQR).
- † Since Q measures the average distance of the quartile points from the median, it is a good index of score density at the middle of the distribution.
- † If the scores in the distribution are packed closely together, the quartiles will be near one another and Q will be small and vice versa.



Interpret the *quartile deviation* in the example above and comment on the distribution of scores in the Biology test

QUARTILE MEASURE OF SKEWNESS

- For a symmetric distribution the median (Q_2) lies exactly half way between the other two quartiles.
- If a distribution is skewed to the right (+ve skew) the median is pulled closer to Q_1 (or pulled closer to Q_3 for -ve skew).
- This relationship enables the derivation of the following coefficient as a measure of skewness.

$$\text{Quartile measure of skewness/ } qsk = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$$



If; $qsk < 0$ - Negatively skewed distribution
 $qsk = 0$ - No skew (normal curve)
 $qsk > 0$ - Positively skewed distribution.

Example

Based on the example above;

$$Q_1 = 14$$

$$Q_3 = 23.43$$

$$Q_2 = 18.39$$

Therefore $qsk = \frac{Q_1 + Q_3 - 2Q_2}{Q_3 - Q_1}$

$$Q_3 - Q_1$$

$$= \frac{14 + 23.43 - 2(18.39)}{23.43 - 14}$$

$$= \frac{0.77}{9.43}$$

$$= 0.08$$

$qsk(0.08) > 0$ hence the distribution is positively skewed.



SELF-TEST

Below are scores in a Chemistry test.

Class	f
50 – 54	2
55 – 59	3
60 – 64	6
65 – 69	9
70 – 74	12
75 – 79	15
80 – 84	10
85 – 89	8
90 – 94	6
95 – 99	4

- i. Compute the mean, mode and median
- ii. Calculate the standard deviation
- iii. Calculate the quartile deviation
- iv. Compute the quartile measure of skewness using an appropriate technique and comment on your answer.

PERCENTILES

- Percentiles are the values of the variate that divide the total frequency into 100 equal parts i.e. the points below which lie 15%, 47%, 82% or any percent of the scores.
- Percentiles are denoted by the symbol P_p , the subscript p refereeing to the percentage of cases below the given value e.g. P_{74} is the point below which lie 74% of the scores.
- Expressed as a percentile, the median is P_{50} while Q_1 is P_{25} and Q_3 is P_{75} . The formula used is as below:

$$P_p = L + \frac{\frac{pN}{100} - F}{f_p} \cdot i$$

Where, P_p = percentage of distribution wanted e.g. 10% percentile, 20% percentile etc.

L = the exact lower limit of the interval in which P_p lies.

pN = part of N to be counted to reach P_p .

F = sum of scores up to L

f_p = the number of scores within the interval in which P_p lies.

i = the width of the classes.

Example

The scores distribution below was obtained in a Biology test. Calculate the 30% percentile and 70% percentile based on the distribution above.

Class	f	cf
0 – 4	2	2
5 – 9	5	7
10 – 14	8	15

15 – 19	9	24
20 – 24	4	28
25 – 29	<u>2</u>	30
N = 30		

Solution

30% percentile of 30 = 9

$$P_p = L + \frac{\frac{pN - F_o}{f_p} \cdot i}{\frac{N - F_o}{f_p} \cdot i}$$

$$P_{30} = 9.5 + \frac{\frac{9 - 7}{8} \cdot 5}{\frac{30 - 7}{8} \cdot 5}$$

$$= 9.5 + 1 \times 5$$

$$= 9.5 + 5$$

$$P_{30} = 14.5$$

70% percentile of 30 = 21

$$P_p =$$

$$P_{70} =$$

$$= 14.5 + 1 \times 5$$

$$= 14.5 + 5$$

$$P_{70} = 19.5$$

Interpretation

30% of the 30 students scored below 14.5 marks while 70% of the 30 students scored below 19.5 marks in the Biology test.

Advantages of percentile

1. Are easy to compute regardless of the shape of the distribution.
2. They are easy to interpret even to lay persons.

Disadvantages

1. They can be assumed to form ordinal scales i.e. the calculations of means and variances of percentiles can produce misleading results leading to inaccurate conclusions.

2. Percentile ranks magnify raw score differences near the middle of the distribution but reduce the raw score differences toward the extreme.

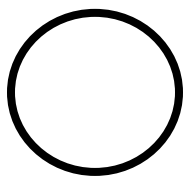


SELF-TEST

The data below relates to weights (in pounds) of refugees in a refugee camp.

Class	f
140 – 144	1
145 – 149	3
150 – 154	2
155 – 159	4
160 – 164	4
165 – 169	6
170 – 174	10
175 – 179	8
180 – 184	5
185 – 189	4
190 – 194	2
195 – 199	1

Calculate the 40%ile and 80%ile based on the distribution above.



Learning Outcomes

You have finished topic 4. The learning outcomes are listed below. Place a (✓) in the column which reflects your understanding.

No.	Learning Outcome	Agree	Disagree
1	I can define the term		
2	I can explain the		

3	I can explain the		
4	I can discuss		

If for whatever reason you have put a tick on any of the statements, go back to the section before you proceed.

However, if you have ticked “agree” on all the statements, you can proceed to the subsequent section



Congratulations! You can continue to the next Topic

Topic 5 MEASURES OF CORRELATION

Introduction

Welcome to this topic on measures of correlation. In the previous topic you were introduced to the measures of variability in which you learnt parameters such as the range, the variance, and the standard deviation that are used to quantify the amount of variation in a set of random variables. In this topic we shall introduce you to various statistical techniques applied in

measures of relationships between two or more data sets. This topic aims to help interpret relationships in students' performance in various tasks given to them.

Topic Objectives



By the end of this topic you should be able to:

- Define correlation
- Illustrate the existence of relationship between the two variables using a scatter diagram
- Compute correlation using product moment correlation coefficient
- Compute correlation using spearman rank order correlation coefficient
- Interpret the kinds of correlation

There are four sections in this topic namely;

Section 1: The concept of correlation analysis

Section 2: Scatter diagram; a graphical presentation of the measures of relationship

Section 3: Spearman and Pearson correlation techniques of determining relationships

Section 4: Regression Analysis

There are questions and activities throughout the topic to help stimulate your thinking. Try to find a quiet place where you can study without being interrupted. In your study you will need a scientific calculator, plain and graph papers for exercises.

We hope you will enjoy reading this topic. We are now ready to start section 1

1.1. The concept of correlation analysis

In this section we will look at the definition and characteristics of correlation analysis

Defining statistical correlation or relationship

In school setting, attributes of the same learner such as academic attainment in various subject fields and the general intellectual ability are observed simultaneously. The observation take the form of scores on tests administered in course of learning may be correlated. Correlating the scores tells us whether the same learner tends to be at about the same level, high. Middle or low on various measures or variables that are correlated.

Statistical correlation is a procedure used to determine the magnitude of the relationship between two sets of scores obtained by a group of test takers in a test or two tests. The correlation analysis involves examining the relationships between variables.

The unit of measure in correlation studies is referred to coefficient of correlation denoted by letter r which stands for the word regression. The concern is to establish the way in which two variables relate to each other for a given group of individuals in classroom, school examinations etc.

Consider the following example:



Do students who join secondary schools with over 400 marks out the possible 500 marks in KCPE score grade B+ and above in KCSE?

Do large classes show lesser gain in knowledge over the year than small classes in secondary schools?

Normally in a relationship, we are concerned with two forms of variables, namely; independent variable and dependent variable. The independent variable influences the dependent variable. The observations for independent variable are denoted X and plotted on the X -axis while the observations for dependent variables are denoted Y and plotted on the Y - axis. This implies that X is the predictor and Y is the predicted. For instant, a student performance in KCPE can be used to predict the student performance in KCSE.

Attributes of correlation coefficients

- The relationship between X and Y with a coefficient of +1.00 indicates a perfect positive correlation. Meaning that X and Y are directly related such that high scores on X are paired with high scores on Y or low scores on X are paired with low scores on Y.
- A correlation of -1.00 indicates a perfect negative relationship or inverse relationship between the variables. This implies high X scores paired with low Y and vice versa. Majority of test takers who scored high in X score low in Y.
- Coefficient of zero indicates complete lack of systematic relationship between the paired scores on X-axis and Y-axis. High X's are likely to be paired with low Y's while low X's are paired with high Y's.
- A correlation between 0.00 and +1.00 or between 0.00 and -1.00 indicates an imperfect relationship. This implies that when the products of X and Y are formed, some will have positive values and others will be negative values.
- A correlation is not expressed as a percentage

1.2. Graphical presentation of the measures of relationship

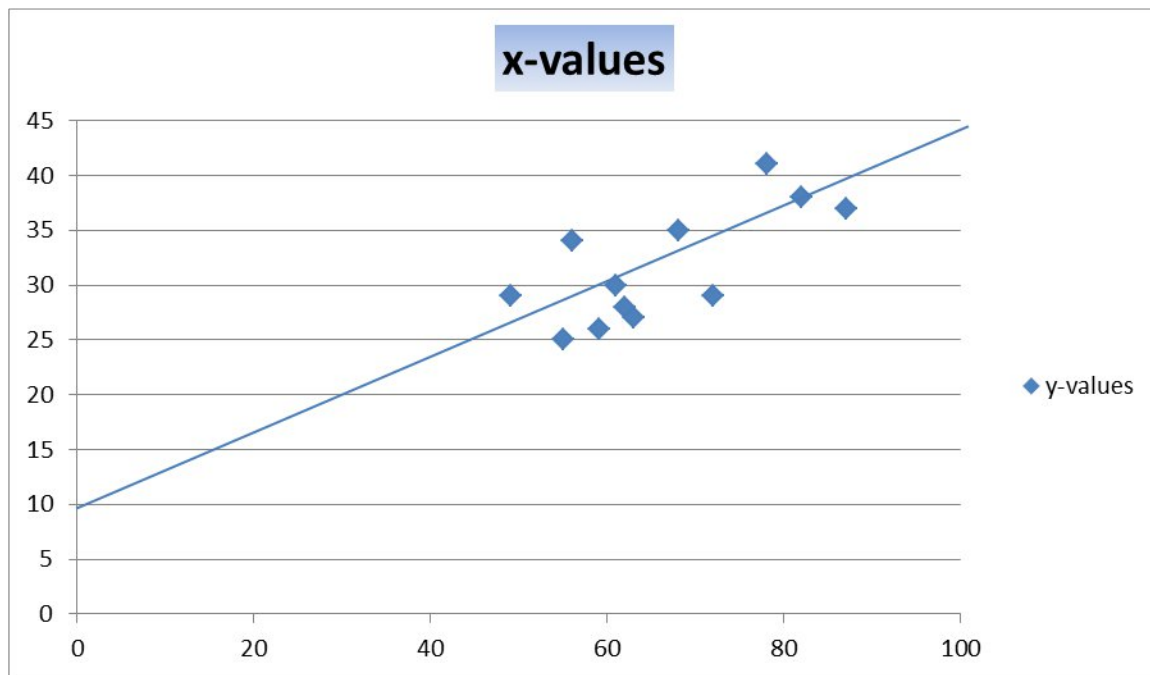
The relationship between the data in the two variables can be presented graphically in a scatter diagram.

Scatter diagram is a graph of data plotted based on two variables where one measure defines the X- axis and the other defines Y- axis. The X and Y values of each individual is represented by a point on the scatter diagram. A mark is placed for each individual at the point of intersection of a straight line perpendicular to X and Y coordinates. A line is drawn through the plotted points on the scatter diagram in a way that it passes through approximately between the patterns of plotted points to determine the kind of relationship between the two variables.

Worked out example: The following data shows performance in math and physics class. Use the scatter diagram to determine the relationship.

Student no.	Math score (x)	Physics score (y)
1	78	41
2	68	35
3	72	29
4	56	34
5	63	27
6	82	38
7	62	28
8	61	30
9	59	26
10	55	25
11	87	37
12	49	29

A scatter diagram showing the relationship between X and Y values



1.3. Methods of determining relationships between variables.

In the previous section, we considered the graphical representation the relationship of data between two variables using the scatter diagram. We are now going to look at how two statistical techniques, namely, spearman rank order correlation coefficient and Pearson product moment correlation coefficient help us to depict relationship between two or more variables.

1.3.1. The Spearman Rank-Order Correlation Coefficient. The spearman Rank Order correlation coefficient is denoted by **rho** or **P** and computed using the formula:

$$Rho \text{ or } P = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

Where;

D = difference/deviation between ranks

n = number of observations

rho is based on ordinal scale with the data ranked from high to low or vice versa. In case of ties ranks are handled by assigning the mean value of ranks to each of the tie holder. Rho is used to determine the measure of internal consistency as well as the measure of stability or reliability of the observations.

Worked out example: The following are the scores obtained in two examinations given to a Kiswahili class.

Exam I	Exam II
50	45
49	50
30	25
11	10
11	15
10	12



Compute the rho and comment on it

Exam I	Exam II	Rank I	Rank II	D	D ²
50	45	1	2	-1	1
49	50	2	1	1	1

30	25	3	3	0	0
11	10	4.5	6	-1.5	2.25
11	15	4.5	4	-0.5	0.25
10	12	5	5	0	0
				$\sum$	4.5

$$r_{h_0/P} = 1 - \frac{\sum D^2}{n(n^2 - 1)} = 1 - \frac{6(4.5)}{6(6^2 - 1)} \quad \text{or} \quad 1 - \frac{6 \times 4.5}{6(6^2 - 1)}$$

$$r_{h_0} = 1 - \frac{27}{210} = 1 - 0.128$$

$$r_{h_0} = 0.877$$

Interpretation

Since rho is strongly/perfectly positive, the scores in the two examinations vary in the same direction. Thus the test is internally consistent or there is positive relationship between the two examinations

Learning Activity 1



The following is the level of patriotism among 10 top leaders in country Z in year 2007 and 2010.

Year 2007	49	50	54	56	59	60	62	61	65	67
Year 2012	21	22	25	34	28	26	30	32	27	31

Compute rho and interpret the result.

Advantages of spearman order correlation

- It is easy to rank the observations
- It is easy to work out the ties by applying mean value calculations.
- The values are small thus easy to work out

Disadvantages

- Where the ties are many it is time wasting to calculate mean values.
- In case of many observations it is laborious working out the rank differences.

1.3.2. Pearson product moment correlation coefficient

To make the required measure of relationship independent of the standard deviation of the two groups of scores, you need to divide $\sum xy$ by $\sum x$ and $\sum y$. The outcome is the measure of relationship between X and Y. This is what is referred to as Pearson product moment correlation coefficient denoted r_{xy} . However, this formula is not ideal for computing r_{xy} . The following two formulas are convenient, namely:

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

where;

r_{xy} the product-moment correlation coefficient

n = the number of scores

$\sum xy$ = the sum of the cross products(each person's x multiplied by his y score

$\sum x \sum y$ = the sum of all the x score multiplied by the sum of all they scores

$\sum x^2$ = the square of each x score added together

$(\sum x)^2$ = the sum of all the x scores, squared

$\sum y^2$ = the square of each y score added together

$(\sum y)^2$ = the sum of all the y scores, squared

or

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Normally the two formulas will yield the same value with very minimal deviation error. r_{xy} never take on a value less than -1 nor a value greater than +1.

r_{xy} is based on an interval scale and the two variables must be similar. The points on the scatter diagram should be uniformly distributed. It provides a linear relationship.



In interpreting correlation between two variable it is incorrect to conclude the one variable causes the other, although this possibility exist. Secondly r_{xy} is not interpreted as percentage; an r_{xy} of 0.75 does not mean 75% relationship and finally r_{xy} is independent of the units in which the two variables are expressed; for example pupils' height in meters can be correlated with their age in years

Steps in computing r_{xy}

Step 1: Add all the raw scores for x; and all the raw scores for y to determine $\sum x$ and $\sum y$

Step 2: Square all x scores and y scores then add the products to determine $\sum x^2$ and $\sum y^2$

Step 3: Multiply x by y for each person then add to determine $\sum xy$

Step 4: Substitute numbers in the formula and perform the necessary operation to determine r_{xy}

Step 5: Interpret the results

Interpretation of r_{xy} values

+1.00 is described as perfect, direct relationship

+.50 is described as moderate, direct relationship

.00 No relationship

-.50 moderate, inverse relationship

-1.00 perfect, inverse relationship

Worked out example

The following scores were obtained by six students of psychology in the two semester examinations. Using r_{xy} . -Determine whether the tests were internally consistent or not.

Candidates	Exam 1 (X)	Exam 2 (Y)	X^2	y^2	XY
C ₁	50	45	2500	2025	2250
C ₂	49	50	2401	2500	2450
C ₃	30	25	900	625	750
C ₄	11	10	121	100	110
C ₅	11	15	121	225	165
C ₆	10	12	100	144	120
Σ	x =161	y =157	$\Sigma x^2 = 6143$	$\Sigma y^2 = 5619$	$\Sigma xy = 5845$

$$\begin{aligned}
 r_{xy} &= \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}} \\
 &= \frac{6 \times 5845 - 161 \times 157}{\sqrt{6 \times 6143 - 25921} \sqrt{6 \times 5619 - 24649}} \\
 &= \frac{35070 - 25277}{\sqrt{10937} \times 9065} \\
 &= \frac{9793}{\sqrt{99143905}} = \frac{9793}{9957.103} \approx 0.9835
 \end{aligned}$$

$$r_{xy} = 0.984 \text{ or } 0.9835$$

Interpretation

There is a strong positive relationship between the 1st semester and second semester examination scores. This means that a student who scored highly in the first semester examinations also scored highly in the second semester examinations. This can also be interpreted to mean that the tests are internally consistent/reliable or that the independent variable (Exam I) has the potential for predicting the dependent variable (Exam II).

By the formula where
$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

candidates	X	Y	X - X	(X - X)2	Y - Y	(Y - Y)2	(X - X)(Y - Y)
C ₁	50	45	23.2	538.24	18.83	354.57	436.86
C ₂	49	50	22.2	492.84	23.83	567.87	529.03
C ₃	30	25	3.2	10.24	-1.17	1.37	-3.74
C ₄	11	10	-15.8	249.64	-16.17	261.47	255.49
C ₅	11	15	-15.8	249.64	-11.17	124.77	176.49
C ₆	10	12	16.8	282.24	-14.17	200.79	-238.07
$\sum$	161	157		1822.84		1510.84	1156.06

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{1159.02}{\sqrt{1822.84 \times 1510.84}} = 1159/1659 = 0.7$$

Learning Activities 2



Out of the two tests you have given to your class in your teaching subject.

- Develop rank order
- Compute r_{xy} using both formulas
- Evaluate the performance of the students in the subject

Summary



In this topic we have learned about the meaning of correlation analysis in which we have looked at the attributes of coefficient of correlations.

We have also looked at the graphical presentation of the measures of relationship using scatter diagram. In addition, we have also learned about methods of determining relationships between variables in which covered Spearman rank order and the Pearson product-moment correlation coefficient.

For example, we have statistically illustrated the relationships between the values of two variables when applying either ρ or r_{xy} and found out that the results are usually within the same range. It is in light of this that we interpret coefficient of correlation to be in the range of +1.00 for perfect positive relationship, 0.00 for no relationship and -1.00 for perfect negative relationship. You are advised to read further and polish your understanding. We hope that you enjoyed reading through this topic.

Suggestions for Further Reading



- Frank S. Freeman (1962). Theory and Practice of Psychological Testing. New Delhi: Mohan Primiani.
- Herert J. Klausmeier & Richard E. Ripple (1971). Learning and Human Abilities; Educational Psychology. New York: Harper and Row Publishers.
- Gene V. Glass & Julian C. Stanley (1970). Statistical Methods in Education and Psychology. New Jersey: Prentice-Hall.

Self-Check 5



The following were marks obtained CAT I and CAT 2 in mathematics by 13 students

CAT 1	24	45	26	30	20	18	54	39	26	44	42	41	22	28
CAT 2	57	49	38	47	17	48	33	39	54	48	50	55	19	50

- i) Compute rho (6 marks)
- ii) Suppose $\sum x = 2354$; $\sum y = 2350$; $\sum x^2 = 132,218$; $\sum y^2 = 139,008$; $\sum xy = 133,538$.
Compute r_{xy} (10 marks)
- iii) Comment on each of the relationships (4 marks)

Scoreboard

Score mark	Remarks
17- 20	Very Good
13-16	Good
8-12	Average
0-7	Below Average



If you have scored a mark of 8 or above congratulations and move to the next topic and if your score is a mark of 7 and below you need to go back and revise the topic thoroughly before you can proceed.

Learning Outcomes

You have now completed topic one, the learning outcome are listed below;

Put a tick in the column which reflects your understanding.

No.	Learning Outcome	Sure	Not Sure
1.	I can now define correlation		
2.	I can well compute scores of two variables using rho		

	and r_{xy}		
3.	I can now interpret the results of coefficient of correlation		

If you have put a tick at the “not sure” column, please go back and study that section in the topic before proceeding.

If you have ticked “sure” in all the rows in all the columns you are ready for the next topic

Congratulations! You can continue to the next topic.

1.4 Regression Analysis

Introduction

Welcome to this topic on regression analysis. In the previous topic you learnt about scatter diagram, spearman rank order and product moment correlation coefficient as statistical techniques for determining variability of two or more variables in a set of data. In this topic we will cover regression analysis as employed in measuring the correlation between two or more data sets.

Topic Objectives



By the end of this topic you should be able to:

- Define regression
- Computer statistical data using regression equation
- Analyze data using simple linear equation
- Determine relationship between variables using the least square regression

This topic consists of six sections namely;

Section 1: The concept of regression analysis

Section 2: Regression equation

Section 3: Simple linear regression

Section 4: The least square regression

I hope you will enjoy reading this topic

2.1. The concept of regression analysis

This is the first section in this topic. In it we shall discuss the concept and the meaning of regression analysis.



What do you understand by the term regression ?

Statistical regression is the brain child of Francis Galton a cousin to Charles Darwin. The term regression refers to the statistical techniques of modeling the relationship between variables. In a cause and effect relationship, the independent variable is the cause, and the dependent variable is the effect. Regression helps to determine the relationship between two variables; an independent variable, denoted by X and a dependent variable, denoted by Y .

2.2. Regression Equation

The regression equation is a linear equation of the form: $\hat{y} = b_0 + b_1x$. To conduct a regression analysis, we need to solve for b_0 and b_1 .

$b_1 = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sum [(x_i - \bar{x})^2]}$	$b_0 = \bar{y} - b_1 * \bar{x}$
--	---------------------------------

Therefore, the regression equation is: $\hat{y} = b_0 + b_1x$.

Worked out example:

In the table below, the x_i column shows scores on a personality test. Similarly, the y_i column shows scores on intelligent test. The last two rows show sums and mean scores that we will use to conduct the regression analysis.

Table1

Student	x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
1	95	85	17	8	289	64	136
2	85	95	7	18	49	324	126
3	80	70	2	-7	4	49	-14
4	70	65	-8	-12	64	144	96
5	60	70	-18	-7	324	49	126
Sum	390	385			730	630	470
Mean	78	77					

Compute the regression equation

$$\hat{y} = b_0 + b_1x .$$

Where:

$b_1 = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sum [(x_i - \bar{x})^2]}$	$b_0 = \bar{y} - b_1 * \bar{x}$
--	---------------------------------

$$b_1 = 470/730 = 0.644$$

$$b_0 = 77 - (0.644)(78) = 26.768$$

Therefore, the regression equation is: $\hat{y} = 26.768 + 0.644x$.

How to Use the Regression Equation

Once you have the regression equation, using it is instant. Choose a value for the independent variable (x), perform the computation, and you have an estimated value ($\hat{y}$) for the dependent variable.

In our example, the independent variable is the student's score on a personality test. The dependent variable is the student's intelligent test. If a student made an 80 on a personality test, the estimated intelligent score would be:

$$\hat{y} = 26.768 + 0.644x = 26.768 + 0.644 * 80 = 26.768 + 51.52 = 78.288$$



When you use a regression equation, do not use values for the independent variable that are outside the range of values used to create the equation. That is called **extrapolation**, and it can produce unreasonable estimates.

In this example, personality test scores used to create the regression equation ranged from 60 to 95. Therefore, only use values inside that range to estimate intelligent score. Using values outside that range (less than 60 or greater than 95) is problematic.

2.3. Methods of computing regression analysis

2.3.1. Simple Linear Regression

When the relationship between the dependent variable (Y) and the independent variable (X) is linear, the technique is called simple linear regression. The simple linear regression model

assume that the relationship between the dependent variable and independent variable can be approximated by a straight line. We can decide whether there is an approximate straight line relationship between Y and X by drawing a scatter diagram of Y versus X. To construct the scatter plot, each value of Y is plotted against its corresponding value of X.

Simple linear regression is appropriated when the dependent variable *Y* has a linear relationship to the independent variable *X*. To check this, make sure that the XY scatter plot is linear.

Linear regression is characterized by two quantities, the slope and Y intercept. These quantities are identified by the coefficients in the equation that describes the linear or a straight line relation between X and Y:

$$Y = a + Bx$$

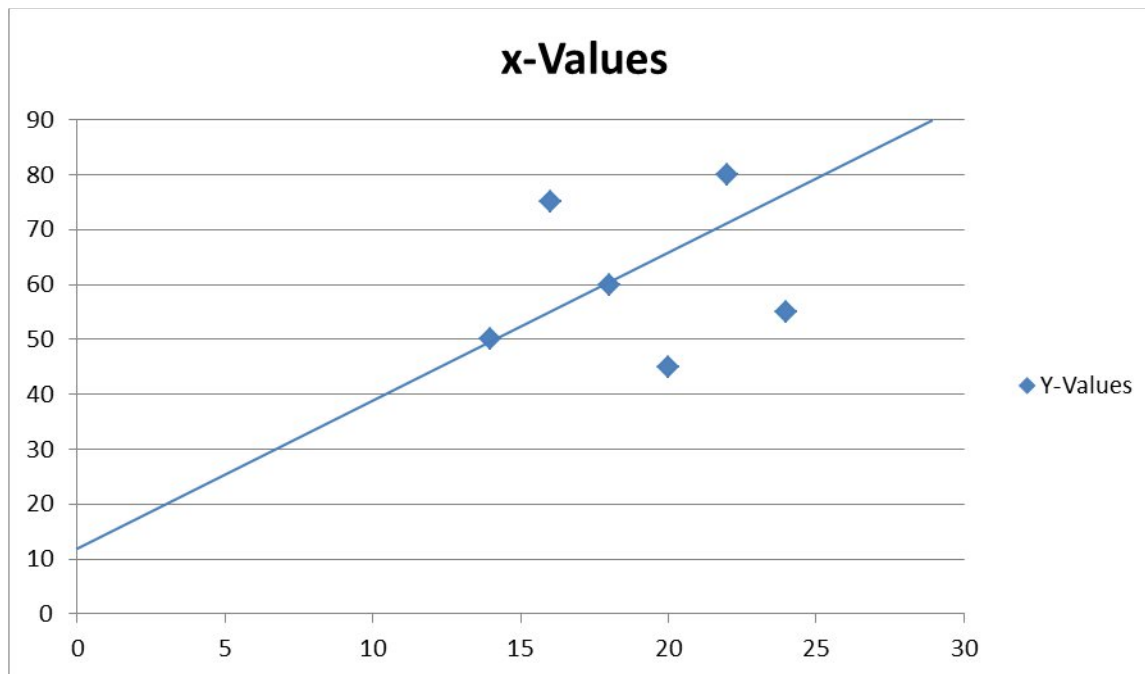
The value **a** is the Y intercept; it measures the level of Y when X is zero. The coefficient **b** is the slope, which gives the change in Y for each unit of change in X.

Worked out example



Suppose we want to determine the relationship between chronological age of learners and their academic performance in class. In this case Age (X) is the predictor of performance (Y).

Age (x)	14	16	18	20	22	24
Performance (y)	50	75	60	45	80	55



With this statistics, scatter diagram is plotted from the pairs of the values of the X and Y variables. From the general pattern of the plotted points on the scatter diagram it is possible to visualize a line that approximates the data in such a case, we can conclude that a linear positive relationship exists between the two variables and a positive (+ve) slope suggests a direct relationship.

Since the points are scattered, this makes it difficult to assume what regression analysis will be. To achieve this approximation, we must **fit** a line to the points in the scatter plot of the data. This involves finding mathematically the slope and Y intercept so that the equation $Y = a + bx$ gives a good representation of the X-Y relation. The easiest method to fit a straight line with freehand sketch though is subjective. To draw this, choose two convenient points that are widely separated to come with a line that fairly approximates the spread of data to give meaningful-Y relationship.

The slope of the regression line, gives the average change in the dependent variable, Y, for each unit change in X. The slope can be either positive or negative, depending on the relationship between X and Y. A positive slope means that for a one-unit increase in X, we can expect an average increase in Y. The slope is negative when there is a decrease in Y values following an increase in X value.

Basic assumption of Simple linear regression

1. Individual values of the dependent variable, Y are statistically independent of one another.
2. For a given x value, there can exist many values of Y. Further, the distribution of possible Y values for any X value is normal.
3. The distribution of possible Y values has equal variance for all values of X.
4. The averages of the dependent variables, Y for all values of the independent variables can be connected to a straight line.

These assumptions can be summarized as follows:

$$Y_i = b_0 + b_1x_i + e_i$$

Where Y_i = value of dependent variable

X_i = Value of the independent value

b_0 = Y - intercept

b_1 = Slope of the regression line

e_i = Error term, or residual (i.e. the difference between the actual Y value and the value of Y predicted by the model)

2.3.2. The method of Least Squares

Freehand sketches as used in simple linear method gives relatively subjective fit to a set of data points. Secondly, the true values of the Y intercept and the slope in the simple linear regression model are unknown. A more objective approach is provided by the method of least squares. With this method, the equation for a straight line is obtained by well-defined calculations. To achieve this we compute a single measure that summarizes the closeness of the fitted line to all the individual points.

Least squares linear regression is a method for predicting the value of a dependent variable Y , based on the value of an independent variable X .

Steps in finding the least square line:

1. Write the general equation for straight line as $\hat{y} = b_0 + b_1x$

where;

$\hat{y}$ = the predicted value of the dependent variable when the value of the independent variable is x .

b_0 = pronounced b zero is the Y intercept,

b_1 = (pronounced b one is the slope of the line,

x is the value of the independent variable,

2 compute the sum of squared errors or deviation, denoted SSE;

Where; $SSE = \sum (y_i - (b_0 + b_1 x_i))^2$

3. To find this line, find the values of the y-intercept b_0 and the slope b_1 that minimize SSE

From table 1, calculate the least squares point estimate

Using the summation compute SSxy and SSxx

$$SS_{xy} = \sum y_i x_i - \frac{(\sum x_i)(\sum y_i)}{N} = 390 \times 385 - \frac{(390)(385)}{5} = 150150 - 30030 = 120120$$

$$SS_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{N} = 31150 - \frac{152100}{5} = 31150 - 30420 = 730$$

It follows the least squares point estimate of the slope b_1 is

$$b_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{120120}{730} = 27.56$$

$$SS_{xx} = 730$$

Because $\hat{y} = \frac{\sum y_i}{5} = \frac{385}{5} = 77$ and $\bar{x} = \frac{\sum x_i}{5} = \frac{390}{5} = 78$

The least squares point estimate of the Y intercept b_0 is

$$b_0 = \hat{y} - b_1 \bar{x} = 77 - (27.56)(78) = 77 - 2149.68 = -2072.68$$

$$\hat{y} = b_0 + b_1 x = 2072.68 + 27.56x$$

Since $b_1 = 2072.68$ is positive we estimate that intelligence increases with thematic apperception

Properties of the Least Squares Regression Line

When the regression parameters (b_0 and b_1) are defined as described above, the regression line has the following properties.

- The line minimizes the sum of squared differences between observed values (the y values) and predicted values (the $\hat{y}$ values computed from the regression equation).
- The least squares line passes through the points $(\bar{X}, \bar{Y})$
- The residuals of all the points in the data set add to zero. This implies that the line lies squarely in the middle of the points in the scatter diagram. This not possible for freehand sketch
- The regression constant (b_0) is equal to the y intercept of the regression line.
- The regression coefficient (b_1) is the average change in the dependent variable (Y) for a 1-unit change in the independent variable (X). It is the slope of the regression line.

Learning Activities



1. Using the two set scores your students obtained in your teaching subject, draw a scatter diagram. On it indicate;
 - i) Compute the least squares regression line
 - ii) Indicate the Y intercept and the slope
 - iii) Comment on the result

Summary



In this topic we have defined regression as the statistical techniques of modeling the relationship between variables and looked at how regression helps to determine the relationship between two variables. We have also looked at the regression equation as a linear equation of the form: $\hat{y} = b_0 + b_1x$. We have also covered simple linear regression and the least squares regression as techniques of computing regression analysis. Assumption of simple linear regression and properties of least square regression have also been discussed. You are advised to revise further worked examples in this topic in order to master the concepts.

Suggestions for Further Reading



Bruce L. Bowerman, Richard T. O'Connell & Michael L. Hand. (2001) Business Statistics in Practice. New Delhi: McGraw-Hill

Daniel Sankowsky (1982) Basic Business Statistics. Ohio: Grid Publishing, Inc.

Frank S. Freeman (1962). Theory and Practice of Psychological Testing. New Delhi: Mohan Primiani.

Philip G. Enns (1985) Basic Statistics; Methods and Applications. Illinois: Richard D. Irwin

Self-Check 2



Five fresher student of engineering were randomly selected to take part in an intelligence test before they began their engineering programme. The engineering department has three questions.

1. What linear regression equation best predicts statistics performance, based on intelligence test scores?(5 marks)

2. If a student made an 80 on the intelligence test, what grade would we expect her to make in

statistics?(10 marks)

3. How well does the regression equation fit the data? (5marks)

Scoreboard

Score mark	Remarks
17- 20	Very Good
13-16	Good
8-12	Average
0-7	Below Average



If you have scored a mark of 8 or above congratulations and move to the next topic and if your score is a mark of 7 and below you need to go back and revise the topic thoroughly before you can proceed.

Learning Outcomes

You have now completed topic five, the learning outcome are listed below;

Put a tick in the columns which reflect your understanding.

No.	Learning Outcome	Sure	Not Sure
4.	I can now give the meaning of regression		
5.	I can well illustrate regression equation		
6.	I can now compute a scatter diagram using least square linear regression		

If you have put a tick at the “not sure” column, please go back and study that section in the topic before proceeding.

If you have ticked “sure” in all the rows in all the columns you are ready for the next topic

Congratulations! You can continue to the next topic.

Topic 6

RELIABILITY AND VALIDITY OF A TEST

Topic 6 consists of the following sections:

Section 1: Validity.....

Section 2: Reliability.....

Section 3: Item Analysis.....

Definition

Test: is a standardized instrument design to measure one or more aspects of personality/behaviour like skill, knowledge, intelligent or aptitude.

RELIABILITY

Reliability is the consistence with which a test measures what it is supposed to measure. It relates to the accuracy and consistency of a test across different forms and conditions.

Reliability co-efficient of a test is computed using the Pearson Product. Moment Correlation Co-efficient (r). Is expressed as the relationship between two repeated measures of the same test to the same subjects under similar conditions.

Types of reliability

- a. Internal consistency/ split-half
- b. Parallel/alternate/comparable forms
- c. Test-retest reliability
- d. Intra marker and inter marker reliability

Internal consistency: indicate the homogeneity of the test in that all the items in the test are assumed to measure the same function or traits. In this method the reliability of the test is determined after a single administration of the test. To achieve internal consistency, split half type of test is used. A single test is split into two sub-tests one comprising the even numbered items and the other second one comprising of the odd numbered items. Each of these tests is half the length of the original test. Each test is scored separately and correlation efficient is computed using scores from both even and odd numbered item sub-tests. Spearman Brown formula is used

to compute the whole test as follows:
$$r_{xx} = \frac{2r_{\frac{1}{2}\frac{1}{2}}}{1 + r_{\frac{1}{2}\frac{1}{2}}}$$

Where, r_{xx} is reliability of the whole test

$r_{\frac{1}{2}\frac{1}{2}}$ is reliability of the half test

Example: Suppose the reliability coefficient of $\frac{1}{2}$ test is 0.70. What will be the reliability coefficient of the whole test?

Solution. $r_{xx} = \frac{2 \times 0.70}{1 + 0.70} = 1.4/1.7$. Therefore $r_{xx} = 0.82$

Parallel/alternate/comparable form: is where two or more parallel or alternate tests are administered to the same group of learners on the same date.

Test-retest: is where a single form of a reasonable test is given twice to the same group within a reasonable time gap like two weeks. Two independent sets of scores are obtained. The two sets are correlated using persons product moment correlation coefficient. It is used to check stability of the test reliability. Low reliability coefficient may be influenced by uncontrolled environmental changes during the second administration, maturation effects, further reading/learning, experience, and memory e.t.c.

Intra and inter marker reliability: intra marker reliability is where the same examiner marking the same responses more than ones generates two sets of scores. Inter marker reliability is where more than one examiner marking the same responses. In both types, a correlation co-efficient is then computed using the obtained scores.

Factors influencing reliability coefficient of test scores.

a) Extrinsic factors:

They are factors that lay outside the test but tend to make the test reliable or unreliable. They are as follows:

1. Group reliability: when group of examinees being tested are homogenous in ability, their reliability coefficient (RC) is likely to be low. However, where the examinees vary widely in their ability (are heterogeneous) the reliability of test score is likely to be higher meaning reliability coefficient of the test is high.

2. Guessing by examinees: guessing by the examinees may raise the total which makes reliability co-efficient superiorly high leading to error variance.

3. Environmental conditions: testing environment need to be conducive e.g. sitting arrangement, noise, aeration, lighting etc. Poor testing environment causes destruction of the mental processes. Panic interferes with memory process. These conditions influences momentary fluctuations in the mindset of the examinee sometimes raising or lowering the scores which affect reliability coefficient of the test.

b) Intrinsic factors:

They are also known as internal factors; those that lay within the test.

1. Length of the test: A test with many items is likely to have high RC as compared to a test with very few items. More number of items increases the potential variability thereby improving the test reliability. Spearman Rank formula is used to calculate RC of the length of a test as follows:

$$\frac{rnn(n)(rtt)}{1 + (n - 1)rtt} \quad \text{Where: } \underline{\hspace{2cm}}$$

rnn is reliability coefficient of the lengthened test

n is the number of times the test has been lengthened

rtt is reliability coefficient of the original test.

Example. A language test with 50 items has a reliability coefficient of 0.78. The test is increased 4 times its present length, what will be its new reliability coefficient.

$$\text{Solution. } rnn = \frac{(4) (.78)}{1 + (4-1).78} = \frac{3.2}{3.4} \quad rnn = 0.94$$

2. Range of the total scores: When the standard deviation of the total score is high RC is also high. And when the standard deviation of the total score is low then the RC is likely to be low.

3. Homogeneity of test items: When test items measure the same function or traits from one item to another then the reliability coefficient will be low.

4. Difficulty in value of test items: When items are too easy or too difficult the test may not give a clear picture of the individual being examined. Items should not be such that they are unanswered or are answered by all examinees, this affect reliability coefficient of the test.

5. Discriminative value: When the test is made by discriminative items. The item total test correlation is likely to be high thus affecting the reliability coefficient positively. Where test

items do not discriminate between the superior and inferior learners, then the total correlation result to low reliability coefficient.

6. Scoring reliability: Scorer reliability means how closely two or more scorers agree in scoring or rating the same set of responses. For example, if they do not agree reliability coefficient is likely to be lowered.

VALIDITY

Validity is the degree to which a test measures what it claims to measure. The validity of the test concerns with what the test measures and what it does so far. For example if a test is designed to measure grammar skills should not test comprehensive skills.

Types of validity

Face validity

This type of validity refers to test validity from the face value (observation) of the test. It is the least important aspect of validity because it needs to be checked through other methods.

Content/curricular validity.

Content validity involves systematic evaluation of the test content to determine whether it covers a representative sample of the subject matter taught. Content validity ensures the subject matter is well covered in the test items and the relevance of the content should be adhered to in the light of the examinees responses to those items.

Criterion-related validity

Refers to how well a test compares with external standards. The items on the test are compared with those of another standardized test. It provides an empirical technique for studying the relationship between the performance on the evaluation instrument (test) and some independent external measure. For example, if an instrument purports to measure performance in a job, the examinee who score high on the instrument must also perform well on the job. There are two types of criterion related validity:

Predictive validity: is concerned with the extent to which a test predicts an individual's performances to specific abilities in future. e.g. K.C.P.E can be used to predict candidate's score in K.C.S.E. In this case, K.C.P.E is the predictor and the K.C.S.E the criterion. If the correlation

is strongly positive the K.C.S.E score vary in the same direction with K.C.P.E scores. This can be computed using PPMCC or Spearman Rank Order

Concurrent validity: indicate the process of validating a new test by correlating it, or otherwise comparing it for agreement, with some present source of information. This source of information might have been obtained shortly before or very shortly after the new test was given. Is the validity used when the test is to distinguish between two or more individuals, whose status at the time of testing is different. This is used to predict the behaviour or performance of individuals presently (not future). For example, it can be used to screen between those students who need remedial learning from those who do not.

Construct validity

Is a measure of the degree to which a score obtained from a test meaningfully and accurately reflects or represents a theoretical concept. A construct indicates hypothesis which tells us that a variety of behaviours will correlate with one another in studies of individual differences and will be similarly affected by experimental treatment e.g. fluency speaking, reading e.t.c

Factors affecting validity of a test

-Length of a test. The longer the test the more reliable and valid it becomes. Homogenous lengthening of the test increases reliability of the test and since validity in homogenous test is dependent on reliability, the test's validity is likely to be high. The length is calculated using the formula:

$$rc(nx) = \frac{(n)(rcx)}{\sqrt{(n+n)(n-1)r_{11}}}$$

Where:

rc is correlation between criterion + test lengthened number of times

rcx is correlation between criterion + test in its original length

n is number of times test is lengthened

r_{11} is reliability coefficient of the test

Example. Suppose a test has a validity coefficient of .5 and a reliability of .4, and it is lengthened 4 times its present length. What would be its new validity?

Formula: $rc(nx) = \frac{(n)(rcx)}{\sqrt{(n+n)(n-1)r_{11}}}$

$$\begin{aligned}
 & \sqrt{(n + n)(n - 1)r11} \\
 rc(nx) &= \frac{(4)(.5)}{\sqrt{4 + 4(4 - 1).4}} \\
 &= \frac{2}{\sqrt{4 + 4(3)(.4)}} \\
 &= \frac{2}{\sqrt{9.6}} = \frac{2}{\sqrt{3.098}} = 0.64557 \text{ or } 0.65
 \end{aligned}$$

-Range of ability. A very limited range of ability of the examinees gives rise to low validity coefficient of the test.

-Ambiguous directions. If the instructions/directions of the test are not clear, the test will be interpreted differently by various examinees. Such items encourage guessing thus lowering the validity of the test.

-Socio-cultural differences. A test developed with a particular culture in mind may not be valid when tested to examinees from other cultures. Differences in socio-economic and cultural practices affect test validity. However, a test that is cross-cultural, the validity will not be affected by cultural differences.

Threats to test validity

-Poor reliability. A test that fails to yield internal consistency and stability is deemed to be invalid

-Response sets. Is the tendency for examinees to give particular specific responses to given questions, e.g. Acquiescence-tendency, this is where an examinee give yes responses to test items or purposively giving/saying no (faking bad).

-Bias cultural or gender bias. Test items may be interpreted differently across different cultures. A test may also be biased when it makes systematic errors in predicting some outcome. e.g. biased towards males versus females.

Hawthorne effect refers to a situation where the examinees' awareness of being in an experimental group may be motivated to perform better than usual due to enthusiasm.

John Henry effect. Is where the examinees in control group strive to perform better when placed in a competitive position with the experimental group e.g JAB vs PSSP students.

The Pygmaton effect is where the examinees endeavour to perform better due to the teachers' expectations and therefore they work harder to meet the teachers' expectations.

Halo effect is the where validity is influenced by the teacher's rating based on previous knowledge about the performance of the examinee. This compromises both internal and external validity e.g performance of student from Alliance high school visa-vi a student from Makhokho high school. Definitely the bias will tend to be towards Alliance student, because is known to perform better nationally.

ITEM ANALYSIS

In section 1 and 2 of this topic we dealt with reliability and validity of measurement and evaluation in reference to school curriculum. In first topic of this module you learnt about setting of tests and examination. This is last section of the topic in which we shall discuss how to analysis items in order to come up with standardized test.



What is item analysis?

Is a statistical technique used for selecting and rejecting item of the test on the basis of their difficulty index and discriminative power. The quality and merit of a test depend upon the individual items of which it is composed. Thus it is absolutely important to analyze each item in the test during the standardization process so as to retain only those items that meet the purpose of the instrument being constructed while poor items are discarded or modified. In item analysis it's important to consider those who performed very well on the total test the (high group) and those who performed most poorly (the low group). The high group should consist of the upper 27% of the total group and the low group the lower 27%.

Purposes of item analysis

1. Select appropriate items for final draft and reject poor ones.
2. To obtain the difficulty index (value) (DV) of all items.
3. Provide the discriminative power (DP) item reliability/validity for differentiating between the capable and less capable examinees.
4. To indicate the functioning of destructors in the multiple-choice items.
5. To provide the basics for preparing the final draft of a test.

Components of item analysis

Two main components considered in evaluating items are;

- difficulty value or index
- discriminative power or value of each item.

Let us discuss each of these components in details

Difficulty value (DV)

Difficulty value is determined by the percentage of individuals able to pass each item. Is denoted DV. The rational of testing is that an item should distinguish among individuals. It should not be so easy that all test takers can pass it; nor should it be so difficult that none is able to pass it. If the item is easy and is answered by everyone is said to have DV 100% or proportion of 1. Such an item should be discarded because it serves no purpose in a test. Item not likely to be answered has DV of 0% or proportion of 0. Thus, should be rejected because it serves no purpose since it cannot be answered correctly by even superior candidates.

The difficulty level of a test item provides some indications of the extent to which the item is doing its job. The power to differentiate between students at different levels is necessary if the test is to have adequate construct validity. Some easy item should be included in the test in order to encourage the student of low ability also some difficult items should be included to challenge the abler students. However, for the purpose of constructing a measuring instrument of maximum quality and usability, most items included should be in the middle range of the difficulty.

The difficulty index is computed by dividing the number of pupil passing the item by the total number of the pupils in the combined high and low group. The formula is:

$$P = R / Nr \text{ where;}$$

P =Difficulty index of the item

R =Number of testees who attempted and answered the item correctly

Nr = Number of testees who attempted the item

Illustration: suppose that an item is passed by 12 of the 16 pupils in the high and 8 of the 16 pupils in the low group. Thus, the item difficulty index is;

$$\text{Difficulty} = 12/16 + 8/16 = 20/32 = .63$$



The smallest possible value of the index is zero and the largest possible value is 1.00; the larger the value, the easier is the item. DV is expressed as a percentage or as a fraction. If the DV tends to 100% then the item was too easy and the vice versa.

Discriminative power (DP)

While the difficulty level of an item determines in part its ability to discriminate between the students of different achievements level. Items of the same difficulty level do not always discriminate equally. Test item should be analyzed to determine whether it discriminates satisfactorily between low and high achievers.

Each item should be analyzed with reference to high, average and low performers. Items should also discriminate between some kind of groupings but not others, depending on the purpose of the test. For example, a test should not favour some socioeconomic group and be unfair to others. DP is used to know who is above or below average in ability. Ideal test items should discriminate sieve between superior and inferior examinees. If item is answered by both superior and inferior candidates or not answered by both groups it should be rejected since it cannot discriminate. If answered correctly by superior and not correctly by inferior examinees then it has high DP thus should be retained because it clearly separates the superior examinee from those who are inferior in the trait/ behavior be measured.

Procedure of calculating DP

Step 1. Sort the test papers into groups based on the total score. The grouping helps to identify the top and the bottom groups

Step 2 Calculate the portion of examinees who get each item correct

Step 3. Calculate the proportion of students in the bottom group who get same item correct

Step 4 Subtract the result in step 3 from results in step 2

Step 5 The resulting figure is the discriminative power.

The following guide according to Nunnally and Berrsten (1994) is used to interpret the discriminative index;

0.40 and above- item is good in discriminating

0.20 -0.40- item is satisfactory in discriminating

Below 0.20- item is poor in discriminating (many weak students get it correct).

0-means both weak and good students answered the item correctly. Thus has no discriminative power.



A test item is good if it can discriminate between the weak and the bright students.

Illustration, suppose that an item is passed by 8 of the 16 pupils in low group. The item will have discriminative power of .50 but clearly would not discriminate between those who did well and those who did poorly on the test as a whole.

On the other hand, suppose that an item is passed, by all of the 16 pupils in the high group and by none of the 16 in the low group; its difficulty index would also be .50 but we would conclude that it had maximum discriminative power.

Dimensions of DP

- a) Positive DP- Is where the % of correct answers is higher with high achievers as compared to lower achievers. i.e item should be accepted.
- b) Negative DP. Is where the % of correct answers is high in the low achievers and low in high achiever. Such items reject.
- c) No discrimination/zero DP. Is where the % of correct answers are equal in both the high and low achievers. Reject since don't make contribution to the function of a test.

A test item is valid if it can discriminate between the weak and good students.

Methods used to determine DP of items

a) Judgment method (short cut way)

Rely on judgment by experts to determine DP. Items are given to group of experts with instructions to give comments. Their comments are incorporated to improve on the reliability and validity of the test. Is equivalent to moderation of a test. Limitation is that experts may be subjective or prejudice the items.

b) Empirical method

Is statistical method where the items are determined on the basis of responses from the respondents/examinees. Secondly is developed based on a portion of responses from the examinees

Learning Activities



1. Sample a set of KCSE national examination and KCSE mock papers in your teaching subjects and validate the difficulty index and discriminative power of each item in the examination paper.
2. Compare the level of difficulty and discriminatory value of the two set of examination papers

Summary



In this section, we looked at meaning of item analysis. We explored various purposes of item analysis. We observed how to compute both the difficulty index and the discriminative power of test item. We went further to discuss guidelines in interpreting and methods used to determine DP

Suggestions for Further Reading



- Frank S. Freeman (1962). Theory and Practice of Psychological Testing. New Delhi: Mohan Primiani.

Gene V. Glass & Julian C. Stanley (1970). Statistical Methods in Education and Psychology. New Jersey: Prentice-Hall.

Richard H. Lindeman (1971). Educational Measurement. New Delhi: Taraporevalia

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Self-Test 3



1. Differentiate between difficulty index and discrimination value of a test item(2 marks)
2. what are the significance of item analysis?(5 marks)
3. What are the attributes of item with reasonable discriminative power?(4 marks)
4. A test has 60 items in which items 55 is answered by only 70 students out of which only 30 of them answered it correctly. What is the difficulty index of item 55? (6 marks)
5. Should the item be retained or discarded. Comment?(3 marks)

Scoreboard

Score mark	Remarks
17- 20	Very Good
13-16	Good
8-12	Average
0-7	Below Average



If you have scored a mark of 8 or above congratulations and move to the next topic and if your score is a mark of 7 and below you need to go back and revise the topic thoroughly before you can proceed.

Learning Outcomes

You have now completed topic one, the learning outcome are listed below;

Put a tick in the column which reflects your understanding.

No.	Learning Outcome	Sure	Not Sure
1.	I can now state the difference between difficulty index and discriminative value of test item		
2.	I can well discuss various of item analysis		
3.	I can now explain the purposes of item analysis		
4.	I can now compute and determine difficulty index of test item		

If you have put a tick at the “not sure” column, please go back and study that section in the topic before proceeding.

If you have ticked “sure” in all the rows in all the columns you are ready for the next topic

Congratulations! You can continue to the next topic.

Answers to self –check

Self-check 5

1. award; formula 1mark, rank table 2marks, substitution 1mark, and result 2marks(total 10marks)
2. award ; formula 2marks, substitutions 6marks, result (0.89) 2marks (total 6 marks)
3. award for each correct interpretation 2marks (total 4marks)

Self-check 2

Self-check 3

Glosary

Bibliography

Bruce L. Bowerman, Richard T. O'Connell & Michael L. Hand. (2001) Business Statistics in Practice. New Delhi: McGraw-Hill

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Formula sheet

$$1. \quad \bar{X} = \frac{\sum x_i}{N}$$

$$2. \quad \bar{X} = \frac{\sum fx}{\sum f} \quad \text{or} \quad \bar{X} = A + \frac{\sum f(x - A)}{\sum f}$$

$$3. \quad Mo = L + \frac{\frac{d_1}{d_1 + d_2}}{\frac{d_1}{d_1 + d_2} + \frac{d_2}{d_1 + d_2}} \cdot \frac{d_1}{d_1 + d_2}$$

$$4. \quad Md = L + \frac{\frac{N/2 - \text{Cum}f_b}{f_w}}{\frac{d_1}{d_1 + d_2} + \frac{d_2}{d_1 + d_2}} \cdot \frac{d_1}{d_1 + d_2}$$

$$5. \quad \text{SIQR} = \frac{Q_3 - Q_1}{2} \quad \text{Where } Q_1 = L + \frac{\frac{N/4 - \text{Cum}f_i}{f_q}}{\frac{d_1}{d_1 + d_2} + \frac{d_2}{d_1 + d_2}} \cdot \frac{d_1}{d_1 + d_2}$$

$$Q_3 = L + \frac{\frac{3N}{4} - \text{Cum}f_i}{f_q}$$

$$6. \quad s^2 = \frac{\sum d^2}{N} \text{ or } s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{N}}{N} \text{ or } s^2 = \frac{\sum f(x - \bar{x})^2}{N}$$

$$7. \quad s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \text{ or } s = \sqrt{\frac{\sum fx^2 - \frac{(\sum fx)^2}{n}}{\sum f}}$$

$$8. \quad r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2 \sum (y - \bar{y})^2]}} \quad \text{or}$$

$$r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$9. \quad \rho / P = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

$$10. \quad b = \frac{N \sum XY - \sum X \sum Y}{N \sum x^2 - (\sum x)^2} \quad a = \bar{Y} - b\bar{X}$$

$$11. \quad Se = S_x \sqrt{1 - r_{xx}}$$

$$12. \quad md = \frac{\sum f|x - \bar{x}|}{\sum f} \quad \text{or} \quad md = \frac{\sum f|x - \bar{x}|}{n}$$

ANSWERS TO SELF TEST

Exercise 1

Solution

X	Tally	f	cf
24	/	1	13
20	//	2	12
19	//	2	10
18	///	3	8
17	////	4	5
15	/	1	1

SELF-TEST 2**Solution**

Step 1: Lowest value = 62; Highest value = 174

$$\backslash (174 - 62) + 1 = 113$$

Step 2: class width = $\frac{113}{10} = 11.3$, rounded off to 11

Step 3: $60 + (11-1) = 69$ class interval is \ 60-69

Step 4: $70 + (11-1) = 79$. Next class interval is 70-79 etc.

Class	Tally	f	cf
60-69	////	4	4
70-79	###	5	9
80-89	### ##	10	19
90-99	### ##	11	30
100- 109	##	5	35
110-119	////	4	39
120 -129	/	<u>1</u>	40

$$\therefore f = 40$$

SELF-TEST 3

Class	f	Class mid points
5 – 9	3	7

10 – 14	4	12
15 – 19	8	17
20 – 24	3	22
25 – 29	2	27

SELF-TEST 4

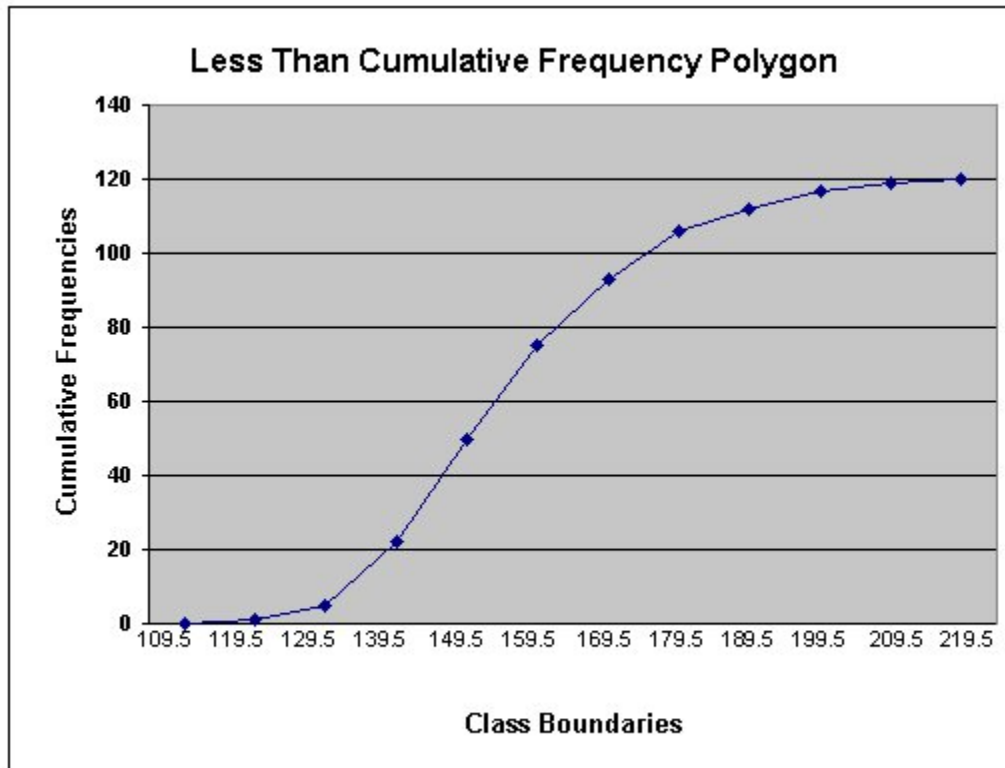
Class	f	True class limits
9-11	1	8.5 - 11.5
12-14	3	11.5 - 14.5
15-17	9	14.5 - 17.5
18-20	14	17.5 - 20.5
21-23	10	20.5 - 23.5
24-26	4	23.5 - 26.5

$$14-1 = \frac{13}{6} = 2.1$$

Self Test 5

Classes	True Class Boundaries	Frequency	Less than Cumulative Frequency	More than Cumulative Frequency
109 – 119	109.5 - 119.5	1	1	119
119 – 129	119.5 - 129.5	4	5	115
129 – 139	129.5 - 139.5	17	22	98
139 – 149	139.5 - 149.5	28	50	70
149 – 159	149.5 - 159.5	25	75	45
159 – 169	159.5 - 169.5	18	93	27
169 – 179	169.5 - 179.5	13	106	14
179 – 189	179.5 - 189.5	6	112	8
189 -199	189.5 - 199.5	5	117	3
199 – 209	199.5 - 209.5	2	119	1
209 – 219	209.5 - 219.5	1	120	0

Sf	Sf	120		
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SELF-TEST 5

Class	Tally	f	cf
20 - 29	////	4	4
30 - 39	### ///	8	12
40 - 49	### ### //	12	24
50 - 59	### ### ### /	16	40
60 - 69	### ### ///	13	53
70 - 79	### //	<u>7</u>	60

$$\therefore f = 60$$



SELF-TEST 8

Calculate the mode of the following data obtained from a Music test among Form three students.

Class	<i>f</i>
30-40	3
40-50	5
50-60	11
60-70	15
70-80	8
80-90	4

$$M_o = L + \frac{\frac{f - f_1}{f_1 + f_2} \cdot i}{\frac{f - f_1}{f_1 + f_2}}$$

$$= 60 + \frac{\frac{4 - 11}{11 + 15} \cdot 10}{\frac{4 - 11}{11 + 15}}$$

$$= 60 + (0.36 \times 10)$$

$$= 60 + 3.636$$

$$M_o = 63.64$$

SELF-TEST TOPIC 3

Compute the mode of the data below using the interpolation formula.

Class	<i>f</i>
20 - 29	4
30 - 39	8
40 - 49	12
50 - 59	16
60 - 69	13
70 - 79	7

$$\therefore f = 60$$

Exercise

Class	<i>f</i>	<i>cf</i>
75-79	3	3
80-84	4	7
85-89	18	25
90-94	20	45
95-99	10	55
100-104	8	63
105-109	5	68
110-114	2	70

$$= 89.5 +$$

$$= 89.5 + (0.5 \times 5)$$

$$= 89.5 + 2.5$$

$$= 92$$

SELF-

<i>Age (yrs)</i>	<i>f</i>	<i>x</i>	<i>fx</i>
20-25	2	22.5	45
25-30	14	27.5	385
30-35	29	32.5	942.5
35-40	43	37.5	1612.5
40-45	33	42.5	1402.5
45-50	<u>9</u>	47.5	<u>56.5</u>
	$\Sigma f = 130$		$\Sigma fx = 4444$

$$\bar{X} = \frac{\Sigma fx}{\Sigma f}$$

$$= \underline{4444}$$

$$130$$

$$= 34.18$$

Variance

X	x^2
10.4	108.16
14.7	216.09
13.6	184.96
14.4	207.36
16.1	259.21
<u>18.5</u>	<u>342.25</u>
$\Sigma x = 87.7$	$\Sigma x^2 = 1318.03$

$$S^2 = \frac{\Sigma x^2}{N} - \frac{(\Sigma X)^2}{N}$$

$$= \frac{1318.03}{6} - \frac{87.7^2}{6}$$

$$= 219.67 - 213.65$$

$$S^2 = 6.02$$

Class	f	x	fx	$X - \bar{X}$	$X - \bar{X}^2$	$f(X - \bar{X})^2$
35-39	3	37	111	-15.17	230.13	690.39
40-44	3	42	84	-10.17	103.43	206.86
45-49	5	47	235	-5.17	26.73	183.65
50-54	8	52	416	-0.17	0.03	0.24
55-59	7	57	399	4.83	23.33	163.31
60-64	3	62	186	9.83	96.63	289.89
65-69	2	67	<u>134</u>	14.83	219.93	<u>439.86</u>
			$\Sigma fx = 1565$			$\Sigma f(X - \bar{X})^2 = 1924.2$

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{1565}{30}$$

$$= 52.17$$

$$S^2 = \frac{\sum f(X - \bar{X})^2}{N}$$

$$= \frac{1924.2}{30}$$

$$S^2 = 64.14$$

Sd

X	$X - \bar{X}$	$(X - \bar{X})^2$
7	-2	4
8	-1	1
9	0	0
9	0	0
9	0	0
10	1	1
11	2	4
$\sum X = 63$		$\sum (X - \bar{X})^2 = 10$

N=7

$$\bar{X} = \frac{\sum x_i}{N}$$

$$\bar{X} - 63 = 9$$

7

$$\begin{aligned}
 &= \sqrt{\frac{\sum (X - \bar{X})^2}{N}} \\
 &= \sqrt{\frac{10}{7}} \\
 &= \sqrt{1.43} \\
 &= 1.19
 \end{aligned}$$

Class	<i>f</i>	<i>x</i>	<i>fx</i>	<i>x</i> ²	<i>f(x</i> ² <i>)</i>
20-24	2	22	44	484	968
25-29	14	27	378	729	10,206
30-34	29	32	928	1024	29,696
35-39	43	37	1591	1369	58,867
40-44	33	42	1386	1764	58,212
45-49	<u>9</u>	47	<u>423</u>	2209	<u>19,881</u>
	Σ <i>f</i> =130		Σ <i>fx</i> = 4750		Σ <i>f(x</i> ² <i>)</i> =177,830

$$\begin{aligned}
 s &= \sqrt{\frac{\sum (x - \bar{x})^2}{\sum f} - \frac{(\sum fx)^2}{(\sum f)^3}} \\
 &= \sqrt{\frac{177,830}{130} - \frac{4750^2}{130^3}}
 \end{aligned}$$

$$= \sqrt{1367.92 - 1335.06}$$

$$= \sqrt{32.86}$$

$$= 5.73$$

<i>Class</i>	<i>f</i>	<i>cf</i>
50 – 54	2	2
55 – 59	3	5
60 – 64	6	11
65 – 69	9	20
70 – 74	12	32
75 – 79	15	47
80 – 84	10	57
85 – 89	8	65
90 – 94	6	71
95 – 99	4	75

$$Md = \frac{L + \left(\frac{N}{2} - Cf_b \right) i}{f_w}$$

$$= 74.5 + \frac{37.5 - 32}{15} \times 5$$

$$= 74.5 + 0.367 \times 5$$

$$= 74.5 + 1.83$$

$$Md = 76.3$$

$$\begin{aligned}
 Q_1 &= L + \frac{\left(\frac{N}{4} - Cumf\right)i}{f_q} \\
 &= 64.5 + \frac{\left(\frac{75}{4} - 11\right)5}{9} \\
 &= 64.5 + 0.86 \times 5 \\
 &= 64.5 + 4.31 \\
 &= 68.81
 \end{aligned}
 \qquad
 \begin{aligned}
 &= 79.5 + 0.925 \times 5 \\
 &= 79.5 + 4.625 \\
 &= 84.13
 \end{aligned}$$

$$\begin{aligned}
 \text{Therefore, } qd/SIQR &= \frac{Q_3 - Q_1}{2} \\
 &= \frac{84.13 - 68.81}{2} \\
 &= \frac{15.32}{2} \\
 &= 7.66
 \end{aligned}$$

Class	<i>f</i>	<i>cf</i>
140 – 144	1	1
145 – 149	3	4
150 – 154	2	6
155 – 159	4	10
160 – 164	4	14
165 – 169	6	20
170 – 174	10	30
175 – 179	8	38
180 – 184	5	43
185 – 189	4	47
190 – 194	2	49
195 – 199	1	50

Calculate the 40%ile and 80%ile based on the distribution above.

- i. 40%ile of 50 = 20
- ii. 80%ile of 50 = 40

$$\begin{aligned}
 40\%ile &= 164.5 + \frac{(20-14)}{6} 5 \\
 &= 164.5 + 5 \\
 &= 169.5
 \end{aligned}$$

$$\begin{aligned}
 80\%ile &= 179.5 + \frac{(43-38)}{5} 5 \\
 &= 179.5 + 5 \\
 &= 184.5
 \end{aligned}$$

Interpretation

40% of the 50 refugees weigh below 169.5 pounds while 80% of the 50 refugees weigh below 184.5 pounds in the sample distribution.